

QUASI-STATIONARY RESPONSE GEOMETRY: THE PERTURBATION CALCULUS AND INFORMATION GEOMETRY OF PARAMETRIZED SUB-MARKOVIAN GENERATORS

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ABSTRACT. Let $\theta \mapsto Q(\theta)$ be a real-analytic family of irreducible Metzler matrices with nonpositive, not identically vanishing row sums — the generators of killed (sub-Markovian) continuous-time Markov chains. The principal spectral datum is the pair (Λ, ν) : the magnitude $\Lambda > 0$ of the spectral abscissa (the *decay rate*) and the associated positive left Perron eigenvector ν , normalized to a probability vector (the *quasi-stationary distribution*, QSD). We develop a closed first-order response calculus for (Λ, ν) on the irreducible locus, whose organizing object is the *rate channel*: the decay rate Λ is a second response coordinate with no analogue in the conservative stationary-chain theory, supplying a rank-one term throughout. Algebraic simplicity and a strict spectral gap are rigid on this locus, giving an invariant-subspace splitting $\mathbb{R}^T = \langle h \rangle \oplus \text{Im}(Q + \Lambda I)$ with rank-one projection $P = h\nu$ and reduced resolvent (group inverse) $R = (Q + \Lambda I)^\#$; the exact derivatives of Λ , ν , and P are a Hellmann–Feynman pairing and a reduced-resolvent action with an explicit normalization correction, reducing to the classical group-inverse sensitivity calculus of conservative chains in the no-killing limit $\Lambda \rightarrow 0$. The QSD map pulls the Fisher–Rao metric of the simplex back to a Fisher information metric $\mathcal{G} = J^\top \text{diag}(1/\nu)J$ — the KL Hessian, with null space the non-identifiable directions — augmented by a rank-one rate channel adjoining $\log \Lambda$. Under a self-consistent coupling, a regular branch approaching a simple fold acquires a rank-one σ^{-1} response of the datum and a σ^{-2} divergence of the pulled-back Fisher metric, gated by an explicit condition fixing *which* channel — the QSD law ν or the rate Λ — inherits it. The four classical pillars (Perron–Frobenius, Kato/group-inverse perturbation theory, information geometry, fold theory) are used as tools; what is new is the killed-case calculus organized around the rate channel, the channel split between $d\nu$ and $d\Lambda$, the which-channel fold-transport bookkeeping with its observable-coupling condition, and an exactly *e*-flat, properly-non-uniform realizability witness.

The calculus transports to the operator setting: under a Foster–Lyapunov drift the principal datum, its fold amplification, and the $\Lambda \rightarrow 0$ recovery lift to non-compact state spaces as a *rigorous corollary of cited QSD theory*, discharged on an exactly-solvable countable birth–death-with-killing instance, with the diffusion case at sketch level. The finite- N fluctuation layer and the resulting fold-sensing Cramér–Rao floor are developed in a companion paper [35], of which this paper is the deterministic core.

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1. INTRODUCTION

1.1. The object of study. Fix a finite set T with $|T| = n \geq 2$. A real $n \times n$ matrix Q is *Metzler* (or *essentially nonnegative*) if its off-diagonal entries are nonnegative, $Q_{ss'} \geq 0$ for $s \neq s'$. Such matrices are exactly the infinitesimal generators of nonnegative one-parameter semigroups $e^{tQ} \geq 0$, and a Metzler matrix with $Q\mathbf{1} \leq 0$ is the generator of a *sub-Markovian* (killed, sub-conservative) continuous-time Markov chain on T : the row defect $q := -Q\mathbf{1} \geq 0$ is the vector of killing rates, and $Q\mathbf{1} = -q \leq 0$ records that probability mass may leak out of T into an adjoined absorbing cemetery. When $q \equiv 0$ the matrix is a conservative generator and the long-run object is a stationary distribution; when $q \not\equiv 0$ the only stationary distribution is the point mass at the cemetery, and the relevant long-run object is instead the *quasi-stationary distribution*.

Concretely, write $\mathfrak{s}(Q) := \max\{\Re \mu \mid \mu \in \text{spec}(Q)\}$ for the *spectral abscissa*. We will see that for an irreducible Metzler Q with $q \not\equiv 0$ the abscissa is a real, algebraically simple, strictly dominant eigenvalue $\mathfrak{s}(Q) = -\Lambda$ with $\Lambda > 0$, carrying strictly positive left and right eigenvectors ν and h . Normalizing $\nu\mathbf{1} = 1$ makes ν a probability vector. The pair

$$(\Lambda, \nu) \in \mathbb{R}_{>0} \times \Delta^\circ, \quad \Delta^\circ := \left\{ p \in \mathbb{R}_{>0}^T \mid \sum_s p_s = 1 \right\},$$

is the *principal spectral datum* of Q . Probabilistically, Λ is the exponential decay rate of survival, $\mathbb{P}(\text{not yet killed at time } t) \asymp e^{-\Lambda t}$, with $1/\Lambda$ the mean time to absorption from ν , and ν is the Yaglom limit, the law of the state conditioned on survival; see Darroch and Seneta [14] for the finite case and [30, 13, 40] for the general theory. None of the geometry below requires the probabilistic reading: it is a statement about the principal spectral datum of an essentially nonnegative matrix, and applies equally to any setting in which such matrices arise parametrically.

1.2. The question. Let Θ be a (finite-dimensional, real-analytic) parameter manifold and $\theta \mapsto Q(\theta)$ a real-analytic family of irreducible Metzler sub-generators. As θ varies, the principal datum $\theta \mapsto (\Lambda(\theta), \nu(\theta))$ traces a curve, surface, or higher-dimensional locus in $\mathbb{R}_{>0} \times \Delta^\circ$. We ask:

- (Q1) *Regularity.* Where is (Λ, ν) analytic in θ , and what obstructs analyticity?
- (Q2) *Response.* What are the exact first-order derivatives $d\Lambda$ and $d\nu$, and what linear-algebraic objects carry them?
- (Q3) *Geometry.* What is the natural Riemannian/statistical geometry induced on Θ by the map $\theta \mapsto (\Lambda(\theta), \nu(\theta))$, and what do its degeneracies mean?
- (Q4) *Stratification.* When the operative parameter is itself constrained by an implicit (self-consistent) relation, how does the geometry behave near the folds of that relation, and how do the folds partition Θ ?

1.3. Results and organization. The paper answers (Q1)–(Q4) by assembling four classical theories into a single response calculus. The lead novelty is small and stated as such: a *rate channel*, the second response coordinate Λ , which has no analogue in the conservative stationary-chain theory and supplies the rank-one terms throughout (§4, §5). Around it the fold result of §6 (Theorem 6.4) is the classical implicit-function blow-up at a simple fold, transported through the chain rule into the (Λ, ν) channels: when an observable of the quasi-stationary datum feeds back into the generator, the closed-loop response of (Λ, ν) inherits the rank-one divergence of the self-consistency branch derivative — at rate σ^{-1} for the datum's response and σ^{-2} for the pulled-back information metric. Its contribution is not the blow-up itself, which is the implicit function theorem near a fold, but the bookkeeping of *which channel sees it*: the QSD channel inherits the divergence *precisely when* $\partial_u \nu$ couples to the fold direction, and the rate channel *precisely when* $\partial_u \Lambda$ does. The sections before it build the machinery this theorem consumes; the sections after it verify the calculus in closed form.

To draw the boundary between new and assembled at the outset: the *assembled* ingredients — Perron–Frobenius theory, Kato perturbation theory, the group-inverse sensitivity calculus of stationary chains, classical information geometry, and the singularity theory of fold maps — are classical and are used here as tools. What we claim as *new* is (a) the killed-case response calculus itself: the group-inverse perturbation formulas for the principal datum of a properly killed generator, organized around the second response coordinate Λ (the rate channel), which has no analogue in the conservative theory and reduces to it as $\Lambda \rightarrow 0$; and (b) the fold amplification theorem with its observable-coupling condition, which transports the implicit-function blow-up through the QSD and rate channels and identifies exactly when each channel sees it. The information-geometry pillar adds (c) an exactly e -flat killed-generator family with properly non-uniform killing and a response image of dimension ≥ 2 (Example 5.10, Proposition 5.11), showing realizability of the flat case by explicit construction. The remaining named statements — the rigidity of simplicity, the channel split, the foliation — are short consequences recorded because later sections consume them, not claims of novelty. The finite- N fluctuation layer and the fold-sensing Cramér–Rao floor, which carry the σ^{-2} amplification of this paper into a statistical resolution limit, are developed in the companion paper [35].

The contributions of the paper, in the order they appear, are the following.

- A closed first-order *response calculus* for the principal datum (Λ, ν) of a killed (sub-Markovian) generator, organized around the rate channel Λ and the reduced resolvent (group inverse) R , reducing to the classical stationary-chain sensitivity theory as $\Lambda \rightarrow 0$ (§4).
- The pulled-back *information geometry* of the QSD map. That $\mathcal{G} = J^\top \text{diag}(1/\nu)J$ is the Fisher information and the KL Hessian is textbook; the new ingredient is the one-line *rate-channel augmentation* adjoining $\log \Lambda$ (Definition 5.5), together with the non-identifiability foliation cut out by the kernel (§5–§6).
- The *fold transport* theorem: the classical implicit-function blow-up at a simple fold, transported through the chain rule into a rank-one singularity — at rate σ^{-1} for the closed-loop datum’s response and σ^{-2} for the information metric — with the contribution being the which-channel-sees-it bookkeeping gated by an explicit observable-coupling condition (§6).
- An *exactly e -flat non-uniform family* (§5): a tilted-reset killed generator whose QSD response image is an exponential family of dimension ≥ 2 *with* properly non-uniform killing ($h \neq \mathbf{1}$, active normalization correction), exercising the killed-case information geometry in the flat, multi-dimensional regime (Example 5.10).
- A record of how the calculus *transports* to the operator setting (§8): under a Foster–Lyapunov drift, QSD existence, the amplification, the group-inverse calculus, and the $\Lambda \rightarrow 0$ recovery lift to non-compact state spaces. This is an *assembly* of cited theorems, not a new one — a rigorous corollary for the countable case, anchored on a worked exactly-solvable instance (Poisson QSD, $\Lambda = \frac{1}{6}$ in the calibrated case), with the diffusion case at sketch level.

The finite- N fluctuation layer and the fold-sensing resolution floor (a Cramér–Rao bound unifying the σ^{-2} amplification with the finite- N width) are the subject of the companion paper [35].

Spectral foundation (§2). Theorem 2.3 establishes that the principal datum is well defined, real-analytic on the irreducible locus Θ_{reg} , and carried by a simple, strictly dominant eigenvalue. Proposition 2.6 records the accompanying rigidity, answering (Q1): simplicity of the Perron eigenvalue cannot fail while irreducibility holds, so the spectral singular set is confined to $\partial\Theta_{\text{reg}}$ — a short consequence of Perron–Frobenius positivity and the continuity of spectra, stated explicitly because the perturbation and geometry sections consume it.

Invariant subspaces and the reduced resolvent (§3). Simplicity gives a canonical splitting $\mathbb{R}^T = \text{Im } P \oplus \ker P$ into Q -invariant subspaces, with $P = h\nu$ the Riesz projection and $R =$

$(Q + \Lambda)^\#$ the group inverse on the complement (Proposition 3.1). This is the algebraic substrate for the response calculus.

Response calculus (§4). Answering (Q2), Theorem 4.1 gives the projection derivative $dP = -(R dQ P + P dQ R)$, the eigenvalue derivative $d\Lambda = -\nu(dQ)h$ (Hellmann–Feynman), and the eigenvector derivative $d\nu = -\nu(dQ)R + (\nu(dQ)R\mathbf{1})\nu$. We give the second-order curvature of Λ (Proposition 4.4) and show the entire calculus reduces, as $\Lambda \rightarrow 0$, to the group-inverse / fundamental-matrix sensitivity theory of stationary Markov chains (Schweitzer [36], Meyer [31], Golub–Meyer [16]), with the correct sign (Theorem 4.6).

Information geometry (§5). Answering (Q3), Theorem 5.2 shows that the QSD map pulls the Fisher–Rao metric of Δ° back to the Fisher information metric $\mathcal{G} = J^\top \text{diag}(1/\nu) J$ of the family $\{\nu(\theta)\}$, equal to the Hessian of Kullback–Leibler divergence, with $\ker \mathcal{G} = \ker d\nu$ identifying the non-identifiable directions. The identity $\mathcal{G} = \text{Fisher} = \text{KL Hessian}$ is textbook (it holds for any smooth statistical family); what is specific to the killed-generator setting is the second response coordinate, so the one new line here is the rank-one *rate-channel augmentation* $\mathcal{G}^\oplus = \mathcal{G} + w d(\log \Lambda) \otimes d(\log \Lambda)$ (Definition 5.5), which adjoins $\log \Lambda$ to the QSD observable. The construction is coordinate-free (Proposition 5.4). We recall that the ambient simplex is dually flat, while the response image inherits the Fisher geometry on constant-rank strata; exact Pythagorean projection identities require additional flatness hypotheses (Remark 5.7), illustrated by the explicit e -flat QSD family of Example 5.8.

Response manifold and stratification (§6). Answering (Q4), Theorem 6.1 exhibits the response image as a low-dimensional immersed submanifold of Δ° on each constant-rank stratum, foliated by non-identifiability leaves. Theorem 6.4 treats self-consistent coupling. The mechanism is classical — along a regular branch approaching a simple fold of the self-consistency relation, the implicit branch derivative $\partial_\theta u^* = -(\partial_u F)^{-1} \partial_\theta F$ has a rank-one σ^{-1} singularity by the implicit function theorem; the chain rule then transports this into the closed-loop datum. The contribution is the *which-channel* bookkeeping: the QSD information metric inherits a rank-one divergence precisely when $\partial_u \nu$ couples to the fold direction, the rate channel precisely when $\partial_u \Lambda$ does (Example 6.5 shows the blow-up is exactly the elementary $d\sqrt{\theta}/d\theta$). Remark 6.6 interprets the discriminant, under properness and Thom–Boardman genericity assumptions, as giving a Whitney-stratified family of regimes; Remark 6.7 points to the corresponding small-noise scaling heuristics (recorded for orientation only).

Examples (§5.1, §6.2, §7). Example 5.8 constructs an exactly e -flat QSD response family, showing that the classical information-projection geometry is realizable inside the killed-generator framework (in the uniform-killing subcase); Example 5.9 complements it with a properly non-uniform family ($h \neq \mathbf{1}$) whose non-identifiability foliation, rank-one metric, and channel split are global, affine, and in closed form; and Example 5.10 combines the two regimes in a *tilted reset*, realizing exact e -flatness *together with* properly non-uniform killing and a response image of dimension ≥ 2 . Example 6.5 gives a closed-form scalar fold verifying the σ^{-2} metric blow-up and the necessity of the observable-coupling condition. Section 7 works the two-state killed chain in closed form, verifies the response formulas and the sign, exhibits the conservative limit, proves the structural *channel split* $\ker d\nu \neq \ker d\Lambda$ via a scaling gauge (Proposition 7.2), and gives a self-consistent three-state example realizing a fold numerically.

All displayed first- and second-order formulas were verified numerically against finite differences on randomly generated irreducible Metzler families; the verification is described in Remark 4.8.

1.4. Relation to the literature. The four pillars are individually classical. Perron–Frobenius theory for nonnegative and Metzler matrices is in Berman–Plemmons [6], Horn–Johnson [20], and Seneta [37]. Analytic perturbation of a simple isolated eigenvalue is Kato [22]; the derivative

of the Perron root of an essentially nonnegative matrix, expressed through the group inverse of the associated M -matrix, is Deutsch–Neumann [15]; and the group-inverse formulation of finite-Markov-chain sensitivity is due to Schweitzer [36], Meyer [31], Campbell–Meyer [10], Golub–Meyer [16], and Cho–Meyer [12]. Information geometry — the Fisher–Rao metric, the ± 1 connections, dual flatness, and the projection theorems — is Čencov [11], Amari–Nagaoka [2], Amari [1], and Ay–Jost–Lê–Schwachhöfer [4]. Whitney stratification and the singularity theory of fold maps are Whitney [42], Mather [29], Thom [38], Golubitsky–Guillemin [17], Arnold [3], and Goresky–MacPherson [18]; the relevant bifurcation normal forms are in Kuznetsov [26] and Guckenheimer–Holmes [19], and the small-noise scaling in van Kampen [41].

The contribution of the present paper is the synthesis: we treat the principal spectral datum of a *killed* (sub-Markovian) generator as a map into $\mathbb{R}_{>0} \times \Delta^\circ$ and show that its differential geometry is governed jointly by the group-inverse perturbation calculus and the pulled-back Fisher–Rao metric, with the non-identifiability foliation and the fold stratification as the manifold-theoretic backbone. The conservative theory of [36, 31] is the boundary case $\Lambda = 0$, and the second, genuinely new response coordinate Λ supplies the rank-one terms throughout. The motivating instance — the response of a metastable population model to parameter interventions — is described in §9.2, but the theory is stated and proved with no reference to any particular model. Concretely, the individual ingredients are classical; the four parts claimed as new — the killed-case group-inverse response calculus organized around the rate channel, the channel split, the fold-transport which-channel bookkeeping, and the ϵ -flat properly-non-uniform realizability witness — are the (a)–(d) stated above (§1), here positioned against the literature rather than re-enumerated; the remaining named statements (Perron rigidity, the non-identifiability foliation) are recorded consequences. The paper may also be read as the methods companion to the modeling application of §9.2, from which the questions (Q1)–(Q4) were abstracted.

Closest to the present perturbation analysis is the work of Rudolf and Wang [34] on the stability of quasi-stationary distributions of killed Markov processes, which gives L^1 -norm bounds on the QSD response to bounded generator perturbations in a self-adjoint operator setting; our concern is complementary — exact first-order derivatives of the principal datum and the information geometry they induce, in the finite irreducible Metzler setting. An information-geometric treatment of a different Markov-chain functional — the class-conditional hitting-time laws of an absorbing chain under a log-linear edge model $A_{ij}(\theta) = A_{ij}^{(0)} e^{\theta^\top \phi_{ij}}$ — was recently given by Kimura [23], who computes the (observed) Fisher information of the hitting-time law, finds it rank-one per seed node, and reads its null space as the non-identifiable edge directions. The structural parallel is real, and it is at the level of the *shared apparatus*: a pulled-back Fisher form that is rank-one with a non-identifiability foliation cut out by its kernel. We claim no novelty for that apparatus — it is classical information geometry, parallel on both sides. The genuine difference is the *object* being parametrized. A first-passage law carries no rate channel; the present paper parametrizes the principal spectral datum (Λ, ν) of a *killed* generator, whose decay rate Λ is a second response coordinate with no analogue in a hitting-time functional (which has no principal eigenvalue datum to differentiate). The increment claimed here is therefore precisely the Perron/resolvent perturbation calculus this datum requires: the group-inverse response formulas of Theorem 4.1, the rate channel and channel split (Proposition 7.2), Perron rigidity on the irreducible killed locus (Proposition 2.6), and the self-consistent fold amplification of Theorem 6.4 (with the saddle-node scaling heuristics of Remark 6.7) — none of which [23] carries, because a first-passage functional has no rate channel to begin with. The non-identifiability geometry is also the “sloppy model” / model-manifold phenomenon [39, 28] of the parameter-estimation literature.

The infinite-dimensional extension of §8 draws on further classical lines: the QSD theory rests on the Champagnat–Villemonais Lyapunov criterion for non-compact killed processes [8, 9],

on Krein–Rutman positivity in a weighted cone [25], and on Meyn–Tweedie Foster–Lyapunov theory [32]. The companion paper [35], which carries the fold amplification into the finite- N and statistical regime, draws in addition on the metastability/critical-slowness literature of Berglund–Gentz [5] and van Kampen [41]. The motivating application — a metastable behavioral population leaking to an absorbing death state — is in the spirit of the opioid-epidemic and post-release-mortality modeling of Bukten et al. [7], though, as emphasized throughout, the theory is stated and proved with no reference to any particular model.

Scope of the results. The core results (§§2–6) are finite-dimensional and are stated on the regular locus of irreducible, properly killed Metzler sub-generators; their operator-theoretic transport to countable and diffusion state spaces, under cited hypotheses, is §8 (a rigorous corollary of cited QSD theory for the countable case, a sketch for the diffusion case). On this locus the Perron datum is analytic and the principal eigenvalue cannot coalesce with the rest of the spectrum; spectral singularities therefore occur only at the boundary, where irreducibility or proper killing fails. The fold results of §6 concern a different source of singular behavior: an implicit self-consistency relation $F(\theta, u) = 0$, whose regular branches may approach a fold of the projection $\mathcal{S} \rightarrow \Theta$ — not a coalescence of the Perron eigenvalue. The Whitney-stratified-regime language and the small-noise fold exponents are used only under the additional genericity/properness or stochastic normal-form assumptions stated there.

Reader’s guide (proof dependencies). Sections 2–7 are the *proved finite-dimensional core*: the Perron–Frobenius foundation (§2), the invariant-subspace splitting and reduced resolvent (§3), the first-order response calculus with its conservative limit (§4), the pulled-back information geometry (§5), the response manifold and fold transport (§6), and the worked examples (§7). Every theorem there is proved in full, and the displayed formulas are independently checked against finite differences (Remark 4.8). Section 8 is operator-theoretic *transport* under cited hypotheses: the countable case is a rigorous corollary of cited QSD theory discharged on an explicit exactly-solvable instance, the diffusion case is at the level of a sketch. Remarks 6.6–6.7 (Whitney stratification, small-noise scalings) are interpretive, recorded under the additional genericity/stochastic hypotheses stated there. The finite- N fluctuation layer and the Cramér–Rao fold-sensing floor are the subject of the companion paper [35]; the present paper is self-contained without them.

2. PRINCIPAL SPECTRAL DATA OF IRREDUCIBLE SUB-GENERATORS

Notation 2.1. Vectors are columns; $\mathbf{1} \in \mathbb{R}^T$ is the all-ones vector; row vectors are written with a left action, e.g. νQ . For a Metzler matrix Q we write $q := -Q\mathbf{1}$ for the *killing vector*. The directed graph $\mathcal{D}(Q)$ has vertex set T and an edge $s \rightarrow s'$ whenever $s \neq s'$ and $Q_{ss'} > 0$; Q is *irreducible* when $\mathcal{D}(Q)$ is strongly connected. We write $\text{spec}(Q)$ for the spectrum and $\mathfrak{s}(Q) = \max\{\Re \mu \mid \mu \in \text{spec}(Q)\}$ for the spectral abscissa.

Definition 2.2 (Sub-generator). A matrix $Q \in \mathbb{R}^{T \times T}$ is a *sub-generator* if it is Metzler ($Q_{ss'} \geq 0$ for $s \neq s'$) and sub-conservative ($Q\mathbf{1} \leq 0$). It is *properly killed* if $q = -Q\mathbf{1} \neq 0$, and *conservative* if $q = 0$. We denote by $\mathcal{M}_{\text{irr}}(T)$ the set of irreducible, properly killed sub-generators on T .

The next theorem is the Perron–Frobenius foundation. It is a routine consequence of the classical theorem applied to a diagonal shift, but we give the full argument because the strict dominance in *real part* (not merely modulus) and the positivity of Λ are exactly the hypotheses the perturbation and geometry sections consume.

Theorem 2.3 (Principal datum). *Let $Q \in \mathcal{M}_{\text{irr}}(T)$. Then:*

- (i) *The spectral abscissa $\mathfrak{s}(Q)$ is an eigenvalue of Q ; it is real, algebraically simple, and strictly dominant: $\Re \mu < \mathfrak{s}(Q)$ for every $\mu \in \text{spec}(Q) \setminus \{\mathfrak{s}(Q)\}$.*

- (ii) The corresponding left and right eigenvectors ν, h may be chosen strictly positive, and are unique up to positive scaling. Normalize $\nu \mathbf{1} = 1$ and $\nu h = 1$.
- (iii) Writing $\mathfrak{s}(Q) = -\Lambda$, one has $\Lambda > 0$ and the identity $\Lambda = \nu q = -\nu Q \mathbf{1}$.
- (iv) (Converse Perron characterization.) $\mathfrak{s}(Q)$ is the unique eigenvalue of Q admitting a strictly positive right eigenvector, and the unique eigenvalue admitting a strictly positive left eigenvector. Equivalently: if $Qh' = \lambda h'$ with $h' > 0$, or $\nu' Q = \lambda \nu'$ with $\nu' > 0$, then $\lambda = \mathfrak{s}(Q)$.

Proof. Choose a scalar $c > \max_s |Q_{ss}|$ and set $M := Q + cI$. Then $M \geq 0$ entrywise: off-diagonals are those of Q (nonnegative), and diagonals are $Q_{ss} + c > 0$. Since a diagonal shift does not change off-diagonal support, $\mathcal{D}(M) = \mathcal{D}(Q)$ is strongly connected, so M is an irreducible nonnegative matrix. By the Perron–Frobenius theorem [20, Thm. 8.4.4] its spectral radius $r := \rho(M) > 0$ is an algebraically simple eigenvalue with strictly positive left and right eigenvectors ν, h , and every $\mu_M \in \text{spec}(M)$ satisfies $|\mu_M| \leq r$.

Because $\text{spec}(Q) = \text{spec}(M) - c$, the value $\mathfrak{s}(Q) = r - c =: -\Lambda$ is an algebraically simple eigenvalue of Q with the same eigenvectors $\nu, h > 0$, proving the eigenvector claims in (ii). For strict dominance, let $\mu = \mu_M - c \in \text{spec}(Q)$ with $\mu \neq -\Lambda$. Then $\Re \mu = \Re \mu_M - c \leq |\mu_M| - c \leq r - c = -\Lambda$. Equality throughout forces $|\mu_M| = r$ and $\Re \mu_M = |\mu_M|$, i.e. μ_M real and nonnegative with modulus r , hence $\mu_M = r$ and $\mu = -\Lambda$, a contradiction. Thus $\Re \mu < -\Lambda = \mathfrak{s}(Q)$, proving (i).

Finally, left-multiply $Q \mathbf{1} = -q$ by the positive row vector ν : $\nu Q \mathbf{1} = -\nu q$. But $\nu Q = -\Lambda \nu$, so $\nu Q \mathbf{1} = -\Lambda(\nu \mathbf{1})$. Hence $\Lambda(\nu \mathbf{1}) = \nu q$. Since $\nu > 0$, $q \geq 0$, and $q \neq 0$, the right side is strictly positive, so $\Lambda > 0$; with $\nu \mathbf{1} = 1$ we get $\Lambda = \nu q$, which is (iii).

For (iv), keep $M = Q + cI \geq 0$ irreducible as above. By Perron–Frobenius the only eigenvalue of M possessing a nonnegative (hence, by irreducibility, strictly positive) eigenvector is the Perron root $\rho(M) = r$. If $Qh' = \lambda h'$ with $h' > 0$, then $Mh' = (\lambda + c)h'$, so $\lambda + c = r$, i.e. $\lambda = r - c = \mathfrak{s}(Q)$. Applying the same to $Q^\top$ (also irreducible Metzler, with $\mathfrak{s}(Q^\top) = \mathfrak{s}(Q)$) gives the left-eigenvector statement, since a strictly positive left eigenvector of Q is a strictly positive right eigenvector of $Q^\top$. $\square$

Definition 2.4 (QSD map). For $Q \in \mathcal{M}_{\text{irr}}(T)$ we call $\nu \in \Delta^\circ$ the *quasi-stationary distribution* and $\Lambda > 0$ the *decay rate*. For a family $\theta \mapsto Q(\theta)$ we call

$$\Phi: \Theta_{\text{reg}} \longrightarrow \Delta^\circ, \quad \Phi(\theta) := \nu(\theta),$$

the *QSD map*, and $\theta \mapsto \Lambda(\theta)$ the *decay-rate function*, where $\Theta_{\text{reg}} := \{\theta \mid Q(\theta) \in \mathcal{M}_{\text{irr}}(T)\}$ is the *regular* (irreducible, properly killed) locus.

Remark 2.5 (Uniform versus non-uniform killing). In the conservative case ($q = 0$), the principal eigenvalue is 0, the right Perron eigenvector is $h = \mathbf{1}$, and the principal datum reduces to the stationary distribution $\nu = \pi$. Proper killing introduces a positive decay rate ($\Lambda > 0$), but it does not by itself force $h \neq \mathbf{1}$. Indeed,

$$h = \mathbf{1} \iff Q \mathbf{1} = -\Lambda \mathbf{1} \iff q = \Lambda \mathbf{1},$$

so the right Perron eigenvector remains constant exactly when killing is uniform across states. In that case the killed chain is a conservative chain thinned by an independent exponential clock: the QSD is the stationary law of $(Q + \Lambda I)$, while Λ records the uniform survival rate. The genuinely new eigenvector geometry appears when the killing is non-uniform. Then $h \neq \mathbf{1}$, the Perron line is no longer the constant direction, and the normalization geometry in the response formulas separates from the conservative stationary-distribution calculus. Thus the killed theory contains both a uniform-killing subcase, where the rate channel is present but decoupled from the QSD eigenvector geometry, and a non-uniform-killing case, where $h \neq \mathbf{1}$ and the normalization correction c_j is no longer identically zero in general.

We now record the answer to (Q1). Real-analyticity is Kato's theorem for a simple isolated eigenvalue; we add the observation — a short consequence of Theorem 2.3 and the continuity of spectra, recorded because everything downstream consumes it — that on Θ_{reg} the eigenvalue is never non-simple, so there is no interior obstruction.

Proposition 2.6 (Analyticity and rigidity). *Let $\theta \mapsto Q(\theta)$ be real-analytic on an open set Θ and valued in the cone of Metzler sub-generators (each $Q(\theta)$ is Metzler with $Q(\theta)\mathbf{1} \leq 0$), and let $\Theta_{\text{reg}} \subseteq \Theta$ be the regular locus. Then Θ_{reg} is open and:*

- (i) $\Theta_{\text{reg}} \ni \theta \mapsto (\Lambda(\theta), \nu(\theta), h(\theta))$ is real-analytic, and so are the spectral projection $P(\theta) = h(\theta)\nu(\theta)$ and the group inverse $R(\theta) = (Q(\theta) + \Lambda(\theta)I)^\#$ of §3.
- (ii) (Rigidity.) *For every $\theta \in \Theta_{\text{reg}}$ the Perron eigenvalue $-\Lambda(\theta)$ is algebraically simple and isolated from the rest of $\text{spec}(Q(\theta))$ by a gap that is bounded below on compact subsets of Θ_{reg} . Consequently the only points at which the principal datum can fail to be analytic, or at which the spectral gap can close, lie on $\partial\Theta_{\text{reg}}$ (where irreducibility or proper killing degenerates).*

Proof. Openness of Θ_{reg} . By hypothesis every $Q(\theta)$ is Metzler and sub-conservative, so within the family Θ_{reg} is cut out by two further conditions, irreducibility and proper killing, and both are open. If $Q(\theta_0)$ is irreducible, $\mathcal{D}(Q(\theta_0))$ is strongly connected through a finite set of strictly positive off-diagonal entries, which stay positive — and, the family being cone-valued, cannot turn negative — under small perturbations; hence $Q(\theta)$ stays irreducible nearby (off-diagonal entries that vanish at θ_0 only add edges and cannot obstruct strong connectedness). The condition $q(\theta) = -Q(\theta)\mathbf{1} \neq 0$ is likewise open. Therefore Θ_{reg} is open.

(ii) By Theorem 2.3(i), at each $\theta \in \Theta_{\text{reg}}$ the eigenvalue $-\Lambda(\theta)$ is algebraically simple and strictly dominant in real part; in particular it is isolated. The gap $\gamma(\theta) := -\Lambda(\theta) - \max\{\Re \mu \mid \mu \in \text{spec}(Q(\theta)), \mu \neq -\Lambda(\theta)\}$ is strictly positive on Θ_{reg} and continuous (eigenvalues depend continuously on the matrix), hence attains a positive minimum on any compact $K \subset \Theta_{\text{reg}}$.

(i) Fix $\theta_0 \in \Theta_{\text{reg}}$. By (ii), $-\Lambda(\theta_0)$ is a simple eigenvalue isolated by a circle γ enclosing it and no other eigenvalue. By Kato's theorem on the analytic perturbation of a finite system of eigenvalues [22, Ch. II, §1.8 and Ch. VII, §1.3], the Riesz projection $P(\theta) = \frac{1}{2\pi i} \oint_\gamma (zI - Q(\theta))^{-1} dz$ is real-analytic near θ_0 (the integrand is analytic jointly in z and θ on $\gamma \times \{\|\theta - \theta_0\| < \epsilon\}$). The simple eigenvalue is $-\Lambda(\theta) = \text{tr}(Q(\theta)P(\theta))$, analytic; its left and right eigenlines vary analytically, and the normalized eigenvectors $\nu(\theta), h(\theta)$ (fixed by $\nu\mathbf{1} = 1, \nu h = 1$, both analytic constraints with non-vanishing Jacobian by simplicity) are analytic. Analyticity of R follows from the resolvent representation $R(\theta) = \frac{1}{2\pi i} \oint_\gamma \frac{(zI - Q(\theta))^{-1}}{z + \Lambda(\theta)} dz$, or directly from $R = (Q + \Lambda I + P)^{-1}(I - P)$ (Proposition 3.1), a composition of analytic maps with invertible argument. $\square$

Remark 2.7. Proposition 2.6(ii) is special to the Metzler/irreducible setting and should be contrasted with general real or complex analytic families, where the dominant eigenvalue can coalesce with a subdominant one along a real-codimension-one (real case) or real-codimension-two (Hermitian case) variety, producing exceptional points where the resolvent has a pole of order > 1 . For irreducible Metzler families no such interior coalescence is possible: Perron–Frobenius forces simplicity and a strict gap. The exceptional set is pushed entirely to the boundary $\partial\Theta_{\text{reg}}$ where the combinatorial structure (irreducibility) or the killing breaks down. We use this in §6 to locate the singular strata.

3. INVARIANT SUBSPACES AND THE REDUCED RESOLVENT

The simplicity of $-\Lambda$ furnishes a canonical pair of complementary Q -invariant subspaces. The whole response calculus of §4 is the differential geometry of how these subspaces rotate with θ , so we set them up carefully.

Proposition 3.1 (Spectral splitting and group inverse). *Let $Q \in \mathcal{M}_{\text{irr}}(T)$ with principal datum (Λ, ν, h) , normalized $\nu h = 1$. Set $A := Q + \Lambda I$ and $P := h\nu$. Then:*

- (i) *P is the spectral (Riesz) projection onto the Perron eigenline: $P^2 = P$, $\text{rank } P = 1$, $\text{Im } P = \langle h \rangle = \ker A$, and $\ker P = \{x \mid \nu x = 0\} = \text{Im } A$. Moreover $PQ = QP = -\Lambda P$.*
- (ii) *The decomposition $\mathbb{R}^T = \text{Im } P \oplus \ker P$ is into Q -invariant subspaces; on $\ker P$ the restriction $A|_{\ker P}$ is invertible.*
- (iii) *A has index 1, and its group inverse $R := A^\#$ is the unique matrix satisfying*

$$AR = RA = I - P, \quad RP = PR = 0. \quad (1)$$

Equivalently $R = (A + P)^{-1}(I - P)$, and R acts as $(A|_{\ker P})^{-1}$ on $\ker P$ and as 0 on $\text{Im } P$. Consequently

$$\nu R = 0, \quad Rh = 0, \quad RQ = QR = RA - \Lambda R = (I - P) - \Lambda R. \quad (2)$$

Proof. (i) $P^2 = h\nu h\nu = h(\nu h)\nu = h\nu = P$ since $\nu h = 1$; $\text{rank } P = 1$ because P is a nonzero outer product. $Ph = h(\nu h) = h$ and for x with $\nu x = 0$, $Px = h(\nu x) = 0$; thus $\text{Im } P = \langle h \rangle$ and $\ker P = \{\nu x = 0\}$. Now $Qh = -\Lambda h$ gives $QP = Qh\nu = -\Lambda h\nu = -\Lambda P$, and $\nu Q = -\Lambda \nu$ gives $PQ = h\nu Q = -\Lambda h\nu = -\Lambda P$. Hence $\langle h \rangle = \ker A$ and, since $-\Lambda$ is a simple (thus semisimple) eigenvalue, $\ker A$ and $\text{Im } A$ are complementary; comparing dimensions, $\ker P = \text{Im } A$.

(ii) Both summands are Q -invariant: $\text{Im } P = \langle h \rangle$ is an eigenline, and $\ker P = \text{Im } A = \text{Im}(Q + \Lambda I)$ is Q -invariant because Q commutes with $Q + \Lambda I$. Simplicity of the zero eigenvalue of A means A restricted to the complementary invariant subspace $\ker P$ is invertible.

(iii) Index 1 is exactly the statement $\ker A \cap \text{Im } A = \{0\}$, i.e. the splitting of (ii). For an index-1 matrix the group inverse exists and is unique with the stated properties [10, Ch. 4]. The formula $R = (A + P)^{-1}(I - P)$ is verified directly: $A + P$ is invertible (it equals A on $\ker P$ and the identity on $\text{Im } P$ up to the eigenvalue, explicitly $(A + P)h = Ah + h = h$ and $(A + P)|_{\ker P} = A|_{\ker P}$ invertible), and the product satisfies (1). The annihilation identities (2) follow: $\nu R = \nu A^\#$, and $\nu A = 0$ together with uniqueness give $\nu R = 0$; similarly $Rh = 0$; and $RQ = R(A - \Lambda I) = RA - \Lambda R = (I - P) - \Lambda R = QR$. $\square$

Definition 3.2 (Reduced resolvent). The matrix $R = (Q + \Lambda I)^\#$ of Proposition 3.1 is the *reduced resolvent* of Q at the principal eigenvalue $-\Lambda$. It is the inverse of $Q + \Lambda I$ on the complementary invariant subspace $\ker P = \text{Im}(Q + \Lambda I)$ and zero on the Perron line $\langle h \rangle$.

Remark 3.3 (Block picture). In any basis adapted to $\mathbb{R}^T = \langle h \rangle \oplus \ker P$, the matrices Q , P , and R are simultaneously block-diagonal:

$$Q \sim \begin{pmatrix} -\Lambda & 0 \\ 0 & Q_- \end{pmatrix}, \quad P \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad R \sim \begin{pmatrix} 0 & 0 \\ 0 & (Q_- + \Lambda I)^{-1} \end{pmatrix},$$

where $Q_- := Q|_{\ker P}$ has $\text{spec}(Q_-) = \text{spec}(Q) \setminus \{-\Lambda\}$, all with real part $< -\Lambda$. The reduced resolvent is the resolvent of the *subdominant block* evaluated at the Perron eigenvalue. The response calculus below is precisely the off-block-diagonal coupling that a perturbation dQ induces between these two invariant subspaces.

4. THE RESPONSE CALCULUS

Throughout this section $\theta \mapsto Q(\theta)$ is real-analytic on Θ_{reg} , $\partial_j := \partial/\partial\theta_j$, and $Q_j := \partial_j Q$. We write the principal datum and its associated objects without their θ -arguments. By Proposition 2.6 everything in sight is analytic.

4.1. Eigenvalue, eigenvector, and projection.

Theorem 4.1 (First-order response). *On Θ_{reg} , with the normalizations $\nu \mathbf{1} = 1$ and $\nu h = 1$:*

(a) (Projection — coordinate-free.)

$$\partial_j P = -(R Q_j P + P Q_j R). \quad (3)$$

(b) (Decay rate — Hellmann–Feynman.)

$$\partial_j \Lambda = -\nu Q_j h. \quad (4)$$

(c) (Quasi-stationary distribution — reduced resolvent.)

$$\partial_j \nu = -\nu Q_j R + c_j \nu, \quad c_j := \nu Q_j R \mathbf{1}, \quad (5)$$

the term $c_j \nu$ being exactly the correction enforcing $(\partial_j \nu) \mathbf{1} = 0$, so that $\partial_j \nu$ is tangent to Δ° .

(d) (Right eigenvector.) *With the dual normalization fixing $\nu h = 1$,*

$$\partial_j h = -R Q_j h + d_j h, \quad d_j = -c_j. \quad (6)$$

The Jacobian $J := d\Phi = [\partial_j \nu_s]_{s,j}$ is uniformly bounded on compact subsets of Θ_{reg} , as are $\nabla \Lambda := [\partial_j \Lambda]_j$ and dP .

Proof. (b) Differentiate $Qh = -\Lambda h$: $Q_j h + Q \partial_j h = -(\partial_j \Lambda)h - \Lambda \partial_j h$. Left-multiply by ν and use $\nu Q = -\Lambda \nu$ (so $\nu Q \partial_j h = -\Lambda \nu \partial_j h$ cancels the last term) and $\nu h = 1$: $\nu Q_j h = -(\partial_j \Lambda)$, i.e. (4).

(c) Differentiate $\nu Q = -\Lambda \nu$ (row form): $(\partial_j \nu)Q + \nu Q_j = -(\partial_j \Lambda)\nu - \Lambda \partial_j \nu$, hence $(\partial_j \nu)(Q + \Lambda I) = -\nu Q_j - (\partial_j \Lambda)\nu$. Right-multiply by R and use $(Q + \Lambda I)R = I - P = I - h\nu$ (Proposition 3.1) and $\nu R = 0$:

$$(\partial_j \nu)(I - h\nu) = -\nu Q_j R \implies \partial_j \nu - ((\partial_j \nu)h)\nu = -\nu Q_j R.$$

Thus $\partial_j \nu = -\nu Q_j R + c_j \nu$ with $c_j := (\partial_j \nu)h$, the undetermined multiple of the left null vector ν of $Q + \Lambda I$. Imposing the normalization $\nu \mathbf{1} = 1$, equivalently $(\partial_j \nu) \mathbf{1} = 0$, and applying $\mathbf{1}$ to $\partial_j \nu = -\nu Q_j R + c_j \nu$ gives $0 = (\partial_j \nu) \mathbf{1} = -\nu Q_j R \mathbf{1} + c_j$ (here $R \mathbf{1} \neq 0$ in general, so the correction does not drop out), whence $c_j = \nu Q_j R \mathbf{1}$, which is (5). (Note the two expressions $c_j = (\partial_j \nu)h = \nu Q_j R \mathbf{1}$ agree, the first being the intrinsic and the second the normalization-determined value.)

(d) The right version of (c): from $(Q + \Lambda I)\partial_j h = -Q_j h - (\partial_j \Lambda)h$, left-multiply by R , use $R(Q + \Lambda I) = I - P$ and $Rh = 0$: $\partial_j h - (\nu \partial_j h)h = -R Q_j h$, so $\partial_j h = -R Q_j h + d_j h$ with $d_j = \nu \partial_j h$. Differentiating $\nu h = 1$ gives $(\partial_j \nu)h + \nu(\partial_j h) = 0$, i.e. $c_j + d_j = 0$, so $d_j = -c_j$.

(a) Combine. With $P = h\nu$, $\partial_j P = (\partial_j h)\nu + h(\partial_j \nu)$. Substitute (c) and (d):

$$\begin{aligned} \partial_j P &= (-R Q_j h + d_j h)\nu + h(-\nu Q_j R + c_j \nu) \\ &= -R Q_j (h\nu) - (h\nu)Q_j R + (c_j + d_j)h\nu = -R Q_j P - P Q_j R, \end{aligned}$$

since $c_j + d_j = 0$. This is (3); note it is manifestly independent of the normalization, as it must be because P is intrinsic.

Boundedness on compacta: ν, h, R are continuous on Θ_{reg} (Proposition 2.6) and Q_j is continuous, so each displayed expression is continuous, hence bounded on any compact subset of Θ_{reg} . $\square$

Remark 4.2 (Reading the formulas through the invariant splitting). Formula (3) is the Kato projection-perturbation formula [22, Ch. II, §2.1]: it is a sum of two terms, $-P Q_j R$ maps $\ker P$ into $\text{Im } P$ and $-R Q_j P$ maps $\text{Im } P$ into $\ker P$, so $\partial_j P$ is purely “off-block-diagonal” in the splitting of Remark 3.3 — it measures the infinitesimal rotation of the Perron line against its spectral complement. The eigenvalue response (4) is the diagonal block, a Rayleigh quotient derivative; the QSD response (5) is the row read-out of $\partial_j P$ normalized to the simplex. The reduced resolvent R , the inverse of the subdominant block, is the single object coupling the two channels.

Remark 4.3 (The intrinsic and the normalized parts). The vector $-\nu Q_j R$ in (5) is intrinsic: it pairs to zero with h , since $Rh = 0$, and therefore lies in the hyperplane $\{y \mid yh = 0\}$ canonically complementary to $\langle \nu \rangle$ in the dual. The correction $c_j \nu$ moves this intrinsic vector into the simplex tangent space $\{y \mid y\mathbf{1} = 0\}$. These two hyperplanes coincide precisely when $h = \mathbf{1}$, equivalently when the killing is uniform, $q = \Lambda \mathbf{1}$. In that case $R\mathbf{1} = Rh = 0$, so $c_j = \nu Q_j R\mathbf{1} = 0$, and the QSD response has the same normalization form as in the conservative stationary theory. Non-uniform killing is what separates the two hyperplanes and makes the correction term $c_j \nu$ generically nonzero — it is no longer identically zero in j , though it can still vanish for a particular direction when $Q_j R\mathbf{1}$ is ν -orthogonal.

4.2. Second-order response. The curvature of the decay-rate function is classical for simple eigenvalues; we record it because it supplies the Hessian of Λ and again exhibits the reduced resolvent. The first and second derivatives of the Perron root of a Metzler matrix through the group inverse of the associated M -matrix are the classical theorem of Deutsch–Neumann [15], of which the eigenvalue responses (4) and (7) are the principal-datum instance; the increment of §4 is the response of the *left*, QSD-normalized eigenvector (5) with its group-inverse correction c_j , the rate channel Λ as a second response coordinate, and their synthesis with the Fisher geometry of §5.

Proposition 4.4 (Curvature of the decay rate). *On Θ_{reg} ,*

$$\partial_i \partial_j \Lambda = -\nu (\partial_i \partial_j Q) h + \nu Q_i R Q_j h + \nu Q_j R Q_i h. \quad (7)$$

In particular, if $\theta \mapsto Q(\theta)$ is affine (so $\partial_i \partial_j Q = 0$), the Hessian of Λ is $\text{Hess } \Lambda = [\nu Q_i R Q_j h + \nu Q_j R Q_i h]_{ij}$, the symmetrized reduced-resolvent quadratic form in the perturbation directions.

Proof. Differentiate (4), $\partial_j \Lambda = -\nu Q_j h$, in θ_i :

$$\partial_i \partial_j \Lambda = -(\partial_i \nu) Q_j h - \nu (\partial_i \partial_j Q) h - \nu Q_j (\partial_i h).$$

Insert $\partial_i \nu = -\nu Q_i R + c_i \nu$ and $\partial_i h = -R Q_i h + d_i h$ with $d_i = -c_i$:

$$-(\partial_i \nu) Q_j h = \nu Q_i R Q_j h - c_i \nu Q_j h, \quad -\nu Q_j (\partial_i h) = \nu Q_j R Q_i h + c_i \nu Q_j h.$$

The two terms $\mp c_i \nu Q_j h$ cancel, leaving (7). $\square$

Remark 4.5. The form (7) need not be sign-definite — R is the resolvent of the subdominant block and carries no positivity — so Λ is in general neither convex nor concave in θ . Sign-definiteness of $\text{Hess } \Lambda$ along a direction is the infinitesimal signature of locally diminishing or increasing returns of that perturbation on the decay rate.

4.3. The conservative limit. We now make precise the sense in which the calculus reduces, as the killing is switched off, to the classical group-inverse / fundamental-matrix sensitivity theory of stationary Markov chains. The point to keep in view is that the killed-case calculus is *not* a new analytic object: it is exactly the conservative group-inverse calculus of Schweitzer [36] and Meyer [31], with two additions — one extra response coordinate (the decay rate Λ , the eigenvalue that is pinned to 0 in the conservative case) and the normalization correction c_j (which vanishes in that case because $Q_0^\# \mathbf{1} = 0$). Setting $\Lambda \rightarrow 0$ removes both and recovers the classical theory verbatim, as the next theorem states.

Theorem 4.6 (No-killing limit). *Let $\theta \mapsto Q(\theta)$ be a real-analytic family with $Q(\theta)$ Metzler and irreducible, and suppose $q(\theta) = -Q(\theta)\mathbf{1} \rightarrow 0$ along a path on which $Q \rightarrow Q_0$ conservative ($Q_0 \mathbf{1} = 0$). Then:*

- (i) $\Lambda \rightarrow 0$, $h \rightarrow \mathbf{1}$, $\nu \rightarrow \pi$, the stationary distribution of Q_0 ($\pi Q_0 = 0$, $\pi \mathbf{1} = 1$);
- (ii) $R \rightarrow Q_0^\#$, the group inverse of the conservative generator, which satisfies $Q_0^\# = -Z$ with $Z := (\mathbf{1}\pi - Q_0)^{-1} - \mathbf{1}\pi$ the fundamental matrix of Schweitzer [36];

(iii) the QSD response (5) reduces to the stationary response

$$\partial_j \pi = -\pi Q_j Q_0^\# = +\pi Q_j Z, \quad (8)$$

the correction term $c_j \nu$ vanishing because $Q_0^\# \mathbf{1} = 0$;

(iv) the Hellmann–Feynman response (4) degenerates to $\partial_j \Lambda \rightarrow -\pi Q_j \mathbf{1} = \pi(\partial_j q)$, the sensitivity of the (now infinitesimal) total killing rate.

Proof. (i) When $q = 0$, $Q_0 \mathbf{1} = 0$ means $\mathbf{1}$ is a right null vector, so $\Lambda = 0$ and $h \propto \mathbf{1}$; the normalization $\nu h = 1$ with $\nu \mathbf{1} = 1$ forces $h = \mathbf{1}$. The left Perron vector ν becomes the stationary distribution π . Continuity of the simple principal datum (Proposition 2.6) gives the limits along the path.

(ii) As $\Lambda \rightarrow 0$ and $h \rightarrow \mathbf{1}$, $A = Q + \Lambda I \rightarrow Q_0$ and $P = h\nu \rightarrow \mathbf{1}\pi$, so by continuity of the group inverse on the index-1 stratum [10, Ch. 4], $R = A^\# \rightarrow Q_0^\#$. The identity $Q_0^\# = -Z$ is Meyer’s [31, Thm. 2.1]: from $Z = (\mathbf{1}\pi - Q_0)^{-1} - \mathbf{1}\pi$ one checks $-Q_0 Z = I - \mathbf{1}\pi$ and $Z\mathbf{1} = 0$, $\pi Z = 0$, which are exactly the group-inverse axioms for $-Q_0^\#$.

(iii) In (5), $c_j = \nu Q_j R \mathbf{1} \rightarrow \pi Q_j Q_0^\# \mathbf{1} = 0$ because $Q_0^\# \mathbf{1} = 0$ (a group-inverse axiom: $Q_0^\# P = 0$ and $P\mathbf{1} = \mathbf{1}$, so $Q_0^\# \mathbf{1} = Q_0^\# P \mathbf{1} = 0$). Hence $\partial_j \pi = -\pi Q_j Q_0^\# = +\pi Q_j Z$.

(iv) Immediate from (4) with $h \rightarrow \mathbf{1}$ and $\partial_j \Lambda = -\nu Q_j h \rightarrow -\pi Q_j \mathbf{1} = \pi \partial_j(-Q\mathbf{1})|_0 = \pi \partial_j q$. $\square$

Remark 4.7 (Sign). The sign in (8) is positive in the convention $\partial_j \pi = +\pi Q_j Z$ (equivalently $-\pi Q_j Q_0^\#$, since $Q_0^\# = -Z$). A one-line check fixes it: for the two-state conservative chain $1 \rightarrow 2$ at rate a , $2 \rightarrow 1$ at rate b , one has $\pi = (b, a)/(a + b)$ and $\partial \pi_1 / \partial a = -b/(a + b)^2$, which is exactly $+\pi Q_a Z$ with $Q_a = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$; the opposite sign gives $+b/(a + b)^2$ and is wrong. We verified (8) numerically on this and on random multi-state chains (Remark 4.8).

Remark 4.8 (Numerical verification). Every displayed first- and second-order identity — (3), (4), (5), (6), (7), the Fisher identity (9) below, the channel split of Proposition 7.2, and the conservative-limit identities of Theorem 4.6 — was checked against central finite differences on randomly generated real-analytic irreducible Metzler families $Q(\theta)$ with killing supported on a single state, at generic operating points θ_0 . Agreement was to relative error $\leq 10^{-8}$ for first derivatives and $\leq 10^{-7}$ for the second-order curvature, consistent with the respective finite-difference truncation orders. These finite-difference checks — here and the 2×10^3 -random-point checks reported in the worked examples (Examples 5.9, 5.10, Proposition 5.11; `verify_identities.py`, `analysis/eflat_nonuniform_qsd.py`) — are *not used in the proofs*; they are reproducibility diagnostics for the displayed formulas, the proofs being the exact algebra of §§3–4.

5. THE INFORMATION GEOMETRY OF THE RESPONSE MAP

We now equip parameter space with the Riemannian geometry induced by the QSD map. The target Δ° is the open probability simplex; its canonical metric is the Fisher–Rao metric.

Definition 5.1 (Fisher–Rao metric). On $\Delta^\circ = \{p \in \mathbb{R}_{>0}^T \mid \sum_s p_s = 1\}$, with tangent space $T_p \Delta^\circ = \{v \in \mathbb{R}^T \mid \sum_s v_s = 0\}$, the *Fisher–Rao metric* is

$$g_{\text{FR},p}(v, v') = \sum_{s \in T} \frac{v_s v'_s}{p_s}.$$

Its geodesic distance is $d_{\text{FR}}(p, p') = 2 \arccos(\sum_s \sqrt{p_s p'_s})$, realized by the isometry $p \mapsto 2(\sqrt{p_s})_s$ onto the radius-2 sphere; the unit-sphere normalization $\frac{1}{4} g_{\text{FR}}$ gives $d = \arccos(\cdots)$. We use the Fisher-information normalization g_{FR} , under which the pullback below is the Fisher information matrix on the nose.

Theorem 5.2 (Information metric of the QSD family). *Let $\Phi: \Theta_{\text{reg}} \rightarrow \Delta^\circ$, $\Phi(\theta) = \nu(\theta)$, be the QSD map, with Jacobian $J = d\Phi = [\partial_j \nu_s]$ and $D := \text{diag}(\nu)$. Define the symmetric 2-tensor $\mathcal{G} := \Phi^* g_{\text{FR}}$ on Θ_{reg} . Then for all i, j ,*

$$\mathcal{G}_{ij} = \sum_{s \in T} \frac{(\partial_i \nu_s)(\partial_j \nu_s)}{\nu_s} = (J^\top D^{-1} J)_{ij} = \partial_i \partial_j D_{\text{KL}}(\nu_{\theta'} \parallel \nu_\theta) \Big|_{\theta'=\theta}, \quad (9)$$

i.e. $\mathcal{G}$ is simultaneously the pullback of the Fisher–Rao metric, the Fisher information matrix of the statistical model $\{\nu_\theta\}_{\theta \in \Theta_{\text{reg}}}$, and the Hessian of Kullback–Leibler divergence at the diagonal. Moreover $\mathcal{G} \succeq 0$ and

$$\ker \mathcal{G}(\theta) = \ker J(\theta) = \ker d\Phi(\theta).$$

Proof. The first equality is the definition of the pullback evaluated on coordinate vectors: $\mathcal{G}_{ij} = g_{\text{FR}, \nu}(\partial_i \nu, \partial_j \nu) = \sum_s (\partial_i \nu_s)(\partial_j \nu_s) / \nu_s$ (well defined since $\nu_s > 0$ on Θ_{reg} by Theorem 2.3, and $\partial_i \nu \in T_\nu \Delta^\circ$ by Theorem 4.1(c)). The middle equality is the matrix form with $D^{-1} = \text{diag}(1/\nu_s)$.

For the KL identity, write $f(\theta') := D_{\text{KL}}(\nu_{\theta'} \parallel \nu_\theta) = \sum_s \nu_s(\theta') [\log \nu_s(\theta') - \log \nu_s(\theta)]$. Differentiating in θ' and using $\sum_s \partial'_i \nu_s = \partial'_i (\sum_s \nu_s) = \partial'_i (1) = 0$,

$$\partial'_i f = \sum_s (\partial'_i \nu_s) [\log \nu_s(\theta') - \log \nu_s(\theta)] + \underbrace{\sum_s \partial'_i \nu_s}_{=0} = \sum_s (\partial'_i \nu_s) [\log \nu_s(\theta') - \log \nu_s(\theta)].$$

Differentiating once more,

$$\partial'_j \partial'_i f = \sum_s (\partial'_j \partial'_i \nu_s) [\log \nu_s(\theta') - \log \nu_s(\theta)] + \sum_s \frac{(\partial'_i \nu_s)(\partial'_j \nu_s)}{\nu_s(\theta')},$$

and evaluating at $\theta' = \theta$ the first sum vanishes (its bracket is 0), leaving

$$\partial'_i \partial'_j f \Big|_{\theta'=\theta} = \sum_s \frac{(\partial_i \nu_s)(\partial_j \nu_s)}{\nu_s},$$

which is (9). This is the standard fact that the Fisher information is the KL Hessian [2, §2.4].

Positive semidefiniteness is clear from $\mathcal{G} = J^\top D^{-1} J$ with $D^{-1} \succ 0$. For the kernel, $v^\top \mathcal{G} v = \|D^{-1/2} J v\|^2 = 0$ iff $J v = 0$ (as $D^{-1/2}$ is invertible), so $\ker \mathcal{G} = \ker J$. $\square$

Remark 5.3. Operationally, $\{\nu_\theta\}$ is the model in which one observes the location $s \in T$ of the chain conditioned on survival; $\mathcal{G}_{ij}$ is the Fisher information a single such observation carries about θ , and $\ker \mathcal{G}$ the directions of θ no amount of such data can resolve.

Proposition 5.4 (Coordinate-freeness). *$\mathcal{G}$ is a genuine $(0, 2)$ -tensor on Θ_{reg} : under a real-analytic reparametrization $\theta = \theta(\tilde{\theta})$ it transforms as $\tilde{\mathcal{G}} = S^\top \mathcal{G} S$ with $S = \partial \theta / \partial \tilde{\theta}$. Consequently its rank, kernel distribution, and the ratios $v^\top \mathcal{G} v$ between metric-normalized directions are intrinsic, and the optimization “maximize $v^\top \mathcal{G} v$ over $\mathcal{C}$ -unit v ” is well-posed for any other intrinsic quadratic cost $\mathcal{C}$.*

Proof. $\mathcal{G} = \Phi^* g_{\text{FR}}$ is a pullback of a fixed tensor on Δ° along the intrinsically defined map Φ ; pullbacks are functorial under composition with diffeomorphisms of the source, giving the stated transformation law. Rank and kernel of a $(0, 2)$ -tensor are basis-independent, and a generalized Rayleigh quotient $v^\top \mathcal{G} v / v^\top \mathcal{C} v$ is invariant under the simultaneous congruence of $\mathcal{G}$ and $\mathcal{C}$. $\square$

Definition 5.5 (Augmented information metric). Treating the full principal datum $(\nu, \log \Lambda)$ as the observable, equip the product manifold $\Delta^\circ \times \mathbb{R}$ with $g_{\text{FR}} \oplus w(dt)^2$, $w > 0$, $t = \log \Lambda$. The pullback along $\theta \mapsto (\nu(\theta), \log \Lambda(\theta))$ is the *augmented information metric*

$$\mathcal{G}^\oplus = \mathcal{G} + w d(\log \Lambda) \otimes d(\log \Lambda), \quad d(\log \Lambda) = \frac{1}{\Lambda} \nabla \Lambda, \quad (10)$$

a rank- ≤ 1 additive correction to $\mathcal{G}$. Its kernel is $\ker \mathcal{G} \cap \ker \nabla \Lambda$, the directions invisible to both the QSD and the decay rate. Both the rank and this kernel are independent of the weight $w > 0$; only the relative scale of the rate channel against the distributional channel depends on w .

Remark 5.6 (Two channels). Definition 5.5 makes precise that there are two response *channels*: the *distributional channel* $\mathcal{G}$, sensing movement of the conditional law ν within Δ° , and the *rate channel* $w d(\log \Lambda)^{\otimes 2}$, sensing movement of the decay rate Λ . They have different kernels in general (Proposition 7.2): a perturbation may move Λ without moving ν or conversely. We emphasize this is a *decomposition*, not an orthogonal splitting: “orthogonal” would presuppose a metric on $T\Theta$, whereas $\mathcal{G}$ and $\nabla \Lambda$ are two distinct functionals whose kernels merely intersect. The augmented metric adds them; it does not claim they are complementary.

Remark 5.7 (Dual flatness, projection, and the special flat case). The ambient simplex Δ° is a dually flat statistical manifold: in mixture coordinates p_s and exponential coordinates $\eta_s = \log(p_s/p_{s_0})$, the Fisher–Rao metric together with the ± 1 Amari connections gives the standard dualistic geometry, with Kullback–Leibler divergence as the canonical divergence. On each constant-rank stratum of the QSD map $\Phi: \Theta_{\text{reg}} \rightarrow \Delta^\circ$, the image $\Phi(\Theta_{\text{reg}})$ is an immersed statistical submodel carrying the induced Fisher metric of Theorem 5.2. Thus fitting a parametrized killed-chain model to empirical conditional data is naturally a problem of information projection onto the response model. For a general response image, however, one should not expect the full exact Pythagorean theorem of information geometry without additional structure: an arbitrary immersed submanifold of Δ° need not be e -flat or m -flat. What holds generally is the local first-order optimality condition: if $\hat{\nu} \in \Phi(\Theta_{\text{reg}})$ is a regular local minimizer of $D_{\text{KL}}(\rho \parallel \nu)$, then the KL residual is Fisher-orthogonal, in the appropriate dualistic sense, to the tangent space $T_{\hat{\nu}}\Phi(\Theta_{\text{reg}})$. Exact Pythagorean decompositions hold in the classical form when the response model has the corresponding flatness property. The next example shows that such flat response models are realizable within the present killed-generator framework: one may construct analytic families of irreducible sub-generators whose QSD response image is an exponential family.

5.1. An exactly e -flat QSD response family. The preceding examples verify the spectral response formulas and the separation between the distributional and rate channels. We now give an example aimed at the information geometry of §5: a killed-generator family whose quasi-stationary response image is exactly an exponential family. We flag at the outset that this construction lands in the *uniform-killing* subcase ($h = 1$), where the specifically killed-case eigenvector geometry is switched off; it exhibits the classical flat information geometry but does not exercise the paper’s killed-case novelty. The examples that do — properly non-uniform killing with an active normalization correction — are Example 5.9 (one-dimensional) and Example 5.10 (which keeps the flatness *and* the non-uniform killing in dimension ≥ 2), both of which follow.

Example 5.8. Let $T = \{1, \dots, n\}$, fix a reference law $\pi \in \Delta^\circ$, and let $f_1, \dots, f_k: T \rightarrow \mathbb{R}$ be observables linearly independent modulo constants. For $\theta \in \mathbb{R}^k$, define

$$\nu_\theta(s) = \frac{\pi_s \exp\left(\sum_{a=1}^k \theta^a f_a(s)\right)}{Z(\theta)}, \quad Z(\theta) = \sum_{r \in T} \pi_r \exp\left(\sum_{a=1}^k \theta^a f_a(r)\right).$$

Let $\Lambda(\theta) > 0$ be any positive real-analytic function and define

$$Q(\theta) = \mathbf{1}\nu_\theta - I - \Lambda(\theta)I,$$

where $\mathbf{1}\nu_\theta$ denotes the rank-one matrix whose every row is ν_θ . Then $Q(\theta)$ is an irreducible Metzler sub-generator: for $i \neq j$,

$$Q_{ij}(\theta) = \nu_\theta(j) > 0,$$

and

$$Q(\theta)\mathbf{1} = \mathbf{1}(\nu_\theta\mathbf{1}) - \mathbf{1} - \Lambda(\theta)\mathbf{1} = -\Lambda(\theta)\mathbf{1}.$$

Moreover,

$$\nu_\theta Q(\theta) = \nu_\theta(\mathbf{1}\nu_\theta - I - \Lambda I) = \nu_\theta - \nu_\theta - \Lambda(\theta)\nu_\theta = -\Lambda(\theta)\nu_\theta.$$

Thus the quasi-stationary distribution of $Q(\theta)$ is exactly ν_θ , and the decay rate is $\Lambda(\theta)$. Indeed, since $\mathbf{1}\nu_\theta$ has eigenvalues 1 (simple) and 0 (multiplicity $n-1$), the spectrum of $Q(\theta)$ is $\{-\Lambda(\theta)\} \cup \{-(1 + \Lambda(\theta))\}^{(n-1)}$; the principal eigenvalue $-\Lambda(\theta)$ is algebraically simple with spectral gap 1, and the right Perron eigenvector is $h = \mathbf{1}$ (uniform killing $q = \Lambda(\theta)\mathbf{1}$), consistent with Remark 2.5. Consequently the QSD response image is the exponential family generated by the sufficient statistics $f_1, \dots, f_k$. Differentiating the log-density gives

$$\partial_a \log \nu_\theta(s) = f_a(s) - \mathbb{E}_{\nu_\theta}[f_a],$$

so the Fisher pullback metric of Theorem 5.2 is

$$G_{ab}(\theta) = \sum_{s \in T} \nu_\theta(s) (f_a(s) - \mathbb{E}_{\nu_\theta}[f_a]) (f_b(s) - \mathbb{E}_{\nu_\theta}[f_b]) = \text{Cov}_{\nu_\theta}(f_a, f_b).$$

This is the usual Fisher information matrix of a finite exponential family. Hence, in this constructed but completely explicit killed-chain family, the response model is e -flat, and the information-projection and generalized Pythagorean statements of the ambient simplex apply in their classical form. The example also clarifies the limitation of Remark 5.7: exact Pythagorean identities are not consequences of constant rank alone, but of additional flatness structure. Generic QSD response images inherit the Fisher metric, while special response images such as this one inherit the full flat information geometry. The rank-one reset generator may be replaced by any irreducible conservative generator K_θ with stationary law ν_θ , setting $Q(\theta) = K_\theta - \Lambda(\theta)I$. The reset choice is used only because it makes the verification immediate.

The reset family, however, lives in the uniform-killing subcase of Remark 2.5: $q = \Lambda(\theta)\mathbf{1}$, $h = \mathbf{1}$, the normalization correction of Remark 4.3 vanishes identically, and the rate channel decouples — precisely the regime in which the specifically killed-case eigenvector geometry is inactive. The next example exercises the complementary, properly non-uniform regime in closed form.

Example 5.9 (Non-uniform killing: explicit foliation and channel split). Let $T = \{1, 2\}$ with rates $a: 1 \rightarrow 2$, $b: 2 \rightarrow 1$ and killing κ from state 1 only, $\theta = (a, b, \kappa) \in (0, \infty)^3$ — the family of §7.1 below. Killing is properly non-uniform: by Example 7.1, $h \propto (1, (a + \kappa - \Lambda)/a)$ is non-constant for every $\kappa > 0$, so the normalization correction c_j of Theorem 4.1 is active. Writing $\nu = (\nu_1, 1 - \nu_1)$, the quadratic

$$\kappa\nu_1^2 - (a + b + \kappa)\nu_1 + b = 0$$

determines $\nu_1 \in (0, 1)$, and $\Lambda = \kappa\nu_1$ (the identity $\Lambda = \nu q$). Implicit differentiation gives

$$d\nu_1 = \frac{-\nu_1 da + (1 - \nu_1)db - \nu_1(1 - \nu_1)d\kappa}{\sqrt{(a + b + \kappa)^2 - 4\kappa b}},$$

so $J = d\nu$ has rank 1 everywhere and the pullback metric of Theorem 5.2 is the rank-one form $\mathcal{G} = d\nu_1 \otimes d\nu_1 / (\nu_1(1 - \nu_1))$. The non-identifiability foliation of Theorem 6.1 is here *global and affine*: the leaf through a point with QSD value ν_1 is the plane

$$\{(a, b, \kappa) \in (0, \infty)^3 : b(1 - \nu_1) = a\nu_1 + \kappa\nu_1(1 - \nu_1)\},$$

on which the entire quasi-stationary distribution is constant. The rate channel does not vanish along the leaves: $d\Lambda = \kappa d\nu_1 + \nu_1 d\kappa$, so on $\ker J$ one has $d\Lambda = \nu_1 d\kappa \neq 0$ unless $d\kappa = 0$ — the channel split of Proposition 7.2, realized here with $h \neq \mathbf{1}$ rather than through the scaling gauge. Consequently the augmented metric $\mathcal{G}^\oplus$ of Definition 5.5 has the one-dimensional kernel

$\{d\nu_1 = 0, d\kappa = 0\}$: of the two non-identifiable directions of $\mathcal{G}$, exactly one survives the rate channel. All displayed formulas were verified numerically against the spectral solver at 2×10^3 random parameter points.

Example 5.10 (Exact e -flatness with non-uniform killing in dimension ≥ 2). Examples 5.8 and 5.9 each switch off half of the information-geometric claim — the first is e -flat but uniformly killed ($h = \mathbf{1}$), the second properly killed but one-dimensional. A single construction realizes both at once: a *tilted reset* generalizing Example 5.8. Fix a reference $\pi \in \Delta^\circ$, observables $f_1, \dots, f_k$ linearly independent modulo constants, and a fixed *non-uniform* killing vector $\kappa \in \mathbb{R}_{\geq 0}^T$ (not a constant multiple of $\mathbf{1}$). For $\theta \in \mathbb{R}^k$ set the exponential family and decay rate

$$\nu_\theta(s) = \frac{\pi_s \exp(\sum_a \theta^a f_a(s))}{Z(\theta)}, \quad \Lambda(\theta) = \langle \nu_\theta, \kappa \rangle \quad (\text{the identity } \Lambda = \nu q),$$

and the *tilted reset distribution* $m_\theta(s) = \nu_\theta(s) (1 + \kappa_s - \Lambda(\theta))$ — a probability vector, since $\sum_s m_\theta(s) = 1 + \langle \nu_\theta, \kappa \rangle - \Lambda = 1$, and positive whenever $\Lambda < 1 + \min_s \kappa_s$. Define

$$Q(\theta) = \mathbf{1} m_\theta - I - \text{diag}(\kappa).$$

Then $Q(\theta)$ is an irreducible Metzler sub-generator with off-diagonals $Q_{ss'} = m_\theta(s') > 0$ and killing $q = -Q\mathbf{1} = \kappa$, properly non-uniform. The tilt is exactly what keeps ν_θ the quasi-stationary distribution under the non-uniform killing:

$$\nu_\theta Q = (\nu_\theta \mathbf{1}) m_\theta - \nu_\theta - \nu_\theta \text{diag}(\kappa) = m_\theta - \nu_\theta - (\kappa_s \nu_\theta(s))_s = -\Lambda(\theta) \nu_\theta.$$

Since $Q(\theta)$ is irreducible Metzler with $Q\mathbf{1} = -\kappa \leq 0$ and $\kappa \neq 0$ (so $Q \in \mathcal{M}_{\text{irr}}(T)$) and $\nu_\theta > 0$ is a left eigenvector for $-\Lambda(\theta)$, Theorem 2.3 (converse Perron characterization, (iv)) identifies $-\Lambda$ as the spectral abscissa, algebraically simple and strictly dominant. (Numerically the non-principal spectrum sits near $-(1 + \kappa_s)$, as expected for a rank-one perturbation of $-I - \text{diag}(\kappa)$; this locates the subdominant block but is not used as a proof step.) The right Perron vector is $h_s \propto (1 + \kappa_s - \Lambda)^{-1}$, *non-constant* precisely because κ is, so $h \neq \mathbf{1}$ and the normalization correction c_j of Theorem 4.1 is active. Meanwhile the response image is the exponential family $\{\nu_\theta\}$, so the pullback metric of Theorem 5.2 is exactly $\mathcal{G}_{ab} = \text{Cov}_{\nu_\theta}(f_a, f_b)$ — the image is e -flat. Example 5.8 is the untilted special case $\kappa = \Lambda\mathbf{1}$, where $m_\theta = \nu_\theta$ and $h = \mathbf{1}$.

For the calibrated instance $T = \{1, 2, 3, 4\}$, π uniform, $f_1 = (0, 1, 0, 1)$, $f_2 = (0, 0, 1, 1)$, $\kappa = (\frac{1}{10}, \frac{1}{5}, \frac{3}{20}, \frac{1}{4})$, the family is dim-2 with image a genuine two-dimensional e -flat submanifold of the three-dimensional simplex; at $\theta = (0.7, -0.5)$ one gets $\Lambda = 0.186$, $\nu = (0.207, 0.416, 0.125, 0.252)$, and $h = (1.091, 0.983, 1.034, 0.937) \neq \mathbf{1}$. All displayed quantities — the exponential-family identity for ν , the dominance of $-\Lambda$, the Metzler/irreducibility of Q , the active c_j , the response formula (5), and the e -flatness $\mathcal{G} = \text{Cov}$ — were verified against the spectral solver at 2×10^3 random parameter points (analysis/eflat_nonuniform_qsd.py: ν to 4×10^{-16} , $\mathcal{G} - \text{Cov}$ to 2×10^{-10} , the response formula to 3×10^{-10} , and $h \neq \mathbf{1}$ with positive margin at every point).

Proposition 5.11 (Realizability of the killed-case information geometry). *Let $\kappa \in \mathbb{R}_{\geq 0}^T$ be a non-uniform killing vector (not a multiple of $\mathbf{1}$); in particular $\kappa \neq 0$, so killing is proper. Let $f_1, \dots, f_k$ be observables independent modulo constants, write the exponential family $\nu_\theta \propto \pi \exp(\sum_a \theta^a f_a)$ and the scalar decay rate $\Lambda(\theta) = \langle \nu_\theta, \kappa \rangle$, and set*

$$\Theta_\kappa := \{\theta \in \mathbb{R}^k : \Lambda(\theta) < 1 + \min_s \kappa_s\},$$

which is open (by real-analyticity of $\theta \mapsto \Lambda(\theta)$) and equals all of $\mathbb{R}^k$ whenever $\text{osc } \kappa := \max_s \kappa_s - \min_s \kappa_s < 1$. For $\theta \in \Theta_\kappa$ the tilted-reset generator $Q(\theta) = \mathbf{1} m_\theta - I - \text{diag}(\kappa)$ of Example 5.10 (with $m_\theta = \nu_\theta (1 + \kappa - \Lambda)$ a row vector) satisfies: (i) $Q(\theta) \in \mathcal{M}_{\text{irr}}(T)$, i.e. Metzler and irreducible with killing $q = -Q\mathbf{1} = \kappa$ (properly non-uniform); (ii) $\nu_\theta Q = -\Lambda(\theta) \nu_\theta$, and by Theorem 2.3 $-\Lambda$

is the spectral abscissa — algebraically simple and strictly dominant — so the principal datum is the exponential family $\{\nu_\theta\}$; (iii) the right Perron vector $h_s \propto (1 + \kappa_s - \Lambda)^{-1}$ is non-constant (precisely because κ is), so $h \neq \mathbf{1}$ and the normalization correction c_j of Theorem 4.1 is active; and (iv) the pullback Fisher metric is $\mathcal{G}_{ab} = \text{Cov}_{\nu_\theta}(f_a, f_b)$, i.e. the response image is exactly e-flat. Consequently the killed-case information geometry is realizable in the genuinely multi-dimensional ($k \geq 2$), properly-killed, flat regime.

Proof. Throughout, $\theta \in \Theta_\kappa$, so $1 + \kappa_s - \Lambda(\theta) > 0$ for every s .

(i). Then $m_\theta(s) = \nu_\theta(s)(1 + \kappa_s - \Lambda(\theta)) > 0$ for all s , so the off-diagonals $Q_{ss'}(\theta) = m_\theta(s') > 0$ make $Q(\theta)$ an irreducible Metzler matrix. Its reset row is a probability vector,

$$\sum_s m_\theta(s) = \sum_s \nu_\theta(s) + \sum_s \nu_\theta(s)\kappa_s - \Lambda(\theta) \sum_s \nu_\theta(s) = 1 + \langle \nu_\theta, \kappa \rangle - \Lambda(\theta) = 1,$$

so $Q(\theta)\mathbf{1} = \mathbf{1}(\sum_s m_\theta(s)) - \mathbf{1} - \kappa = -\kappa$. Since $\kappa \geq 0$ and $\kappa \neq 0$, $Q(\theta)\mathbf{1} = -\kappa \leq 0$ with $\kappa \neq 0$; hence $Q(\theta) \in \mathcal{M}_{\text{irr}}(T)$ and Theorem 2.3 applies.

(ii). The left eigenrelation is the computation of Example 5.10:

$$\nu_\theta Q = \nu_\theta(\mathbf{1}m_\theta) - \nu_\theta - \nu_\theta \text{diag}(\kappa) = m_\theta - \nu_\theta - (\kappa_s \nu_\theta(s))_s,$$

whose s -component is $\nu_\theta(s)(1 + \kappa_s - \Lambda) - \nu_\theta(s) - \kappa_s \nu_\theta(s) = -\Lambda(\theta) \nu_\theta(s)$; thus $\nu_\theta Q = -\Lambda(\theta) \nu_\theta$. Because $Q(\theta) \in \mathcal{M}_{\text{irr}}(T)$ and $\nu_\theta > 0$, the converse Perron characterization Theorem 2.3(iv) identifies $-\Lambda(\theta) = \mathfrak{s}(Q)$, algebraically simple and strictly dominant. (Self-contained pairing: if $h_* > 0$ is a right eigenvector for the abscissa $\mathfrak{s}(Q) = -\Lambda_*$, then $-\Lambda(\theta)(\nu_\theta h_*) = \nu_\theta Q h_* = -\Lambda_*(\nu_\theta h_*)$ and $\nu_\theta h_* > 0$ force $\Lambda(\theta) = \Lambda_*$.) The principal datum is therefore the exponential family $\{\nu_\theta\}$.

(iii). Take $h_s = C/(1 + \kappa_s - \Lambda(\theta))$ with $C > 0$ fixed by $\nu_\theta h = 1$. Then $m_\theta h = C \sum_s \nu_\theta(s) = C$, so $(Qh)_s = C - (1 + \kappa_s)h_s = -\Lambda(\theta)h_s$, confirming h is the right Perron vector; as $C = (1 + \kappa_s - \Lambda)h_s$, h is non-constant exactly because κ is. Hence $h \neq \mathbf{1}$ and the correction c_j of Theorem 4.1 is active.

(iv). The response image is the exponential family $\{\nu_\theta\}$ with sufficient statistics $f_1, \dots, f_k$, so by Theorem 5.2 the pullback metric is $\mathcal{G}_{ab} = \text{Cov}_{\nu_\theta}(f_a, f_b)$.

Finally, if $\text{osc } \kappa < 1$ then $\Lambda(\theta) = \langle \nu_\theta, \kappa \rangle \leq \max_s \kappa_s < 1 + \min_s \kappa_s$ for every θ , so $\Theta_\kappa = \mathbb{R}^k$. The calibrated $k = 2$ instance and the 2×10^3 -point numerical check are in `analysis/eflat_nonuniform_qsd.py`. $\square$

Remark 5.12 (Flatness and non-uniform killing). The three examples chart the boundary drawn in Remark 2.5. Example 5.8 realizes exact e-flatness but in the uniform-killing subcase ($h = \mathbf{1}$), where the specifically killed-case eigenvector geometry is inactive; Example 5.9 exercises the non-uniform regime ($h \neq \mathbf{1}$, $c_j \neq 0$) — kernel, foliation, and channel split in closed form — but on a one-dimensional image, where flatness is vacuous; and Example 5.10 closes the remaining corner, realizing exact e-flatness *together with* properly non-uniform killing ($h \neq \mathbf{1}$, c_j active) and a response image of dimension ≥ 2 . Thus the killed-case information geometry is realizable in the genuinely multi-dimensional, flat regime, not only in the two degenerate slices — realizability in this regime is shown by the explicit tilted-reset construction (Proposition 5.11).

6. THE RESPONSE MANIFOLD AND ITS SINGULAR STRATA

We now answer (Q4): the shape of the response image as a low-dimensional manifold, the non-identifiability foliation, and the behavior under a self-consistent coupling that introduces folds.

6.1. Constant rank, immersed image, and the non-identifiability foliation.

Theorem 6.1 (Response manifold and foliation). *Let $U \subseteq \Theta_{\text{reg}}$ be a connected open set on which $\text{rank } J \equiv r$ is constant (equivalently $\text{rank } \mathcal{G} \equiv r$, by Theorem 5.2). Then:*

- (i) (Image.) *Each $\theta \in U$ has a neighborhood whose image $\Phi(\cdot)$ is an embedded r -dimensional real-analytic submanifold of Δ° ; its tangent space at $\nu(\theta)$ is $\text{Im } J(\theta) \subseteq T_\nu \Delta^\circ$, on which $\mathcal{G}$ restricts to a Riemannian metric (the induced Fisher–Rao metric of the response model).*
- (ii) (Foliation.) *The distribution $\theta \mapsto \ker J(\theta)$ is integrable, and its leaves — the connected components of the fibers $\Phi^{-1}(\Phi(\theta))$ — form a real-analytic foliation of U by $(\dim \Theta - r)$ -dimensional submanifolds, the non-identifiability leaves. Along a leaf the entire quasi-stationary distribution ν is constant to all orders; $\ker \mathcal{G}$ is the tangent distribution to the foliation.*

Proof. Both statements are the constant-rank theorem [27, Thm. 4.12] applied to the real-analytic map Φ . Constant rank r gives local coordinates in which Φ is the linear projection $(x_1, \dots, x_r, x_{r+1}, \dots) \mapsto (x_1, \dots, x_r, 0, \dots)$; the image is the embedded slice $\{x_{r+1} = \dots = 0\}$ (i), and the fibers are the slices $\{x_1 = \dots = x_r = \text{const}\}$, whose tangent distribution is $\ker d\Phi = \ker J$, integrable with these slices as leaves (ii). The tangent space of the image is $\text{Im } J$, and $\ker \mathcal{G} = \ker J$ by Theorem 5.2. Real-analyticity of the chart follows from the analytic constant-rank theorem since Φ is analytic on Θ_{reg} . $\square$

Remark 6.2 (Identifiability dictionary). $\ker J = \ker \mathcal{G}$ is the set of infinitesimal parameter changes that leave the quasi-stationary distribution fixed — the local *non-identifiabilities* of the family. Theorem 6.1 says these organize into leaves of a foliation on each constant-rank region, so non-identifiability is not pointwise noise but a global geometric structure: one can move along a leaf indefinitely without changing ν . Symmetries of the family (§7) produce explicit leaves. The augmented kernel $\ker \mathcal{G} \cap \ker \nabla \Lambda$ refines this to the changes invisible to both ν and Λ .

6.2. Self-consistent coupling and fold divergence. In many settings the operative parameter is not free but constrained by an implicit equation — a self-consistency or fixed-point relation coupling the generator to a scalar or vector *order parameter* u . We formalize this and show that regular branches approaching a fold have a rank-one implicit-response singularity, which is inherited by the QSD channel when $\partial_u \nu e \neq 0$ and by the decay-rate channel when $\partial_u \Lambda e \neq 0$. The resulting divergence of the pulled-back Fisher metric at the fold is the QSD instance of a broadly observed phenomenon: in statistical physics the Fisher–Rao metric of a parametrized family is singular at a critical point, and this singularity has been used to detect phase transitions without an a priori order parameter [21, 33]. Here the critical point is the fold of the self-consistency relation, and the divergence is rank-one with the explicit which-channel bookkeeping below.

Assumption 6.3 (Self-consistent solution manifold and simple folds). Let $\Theta \subseteq \mathbb{R}^d$ and $U \subseteq \mathbb{R}^m$ be open, let $(\theta, u) \mapsto Q(\theta, u) \in \mathcal{M}_{\text{irr}}(T)$ and $F: \Theta \times U \rightarrow \mathbb{R}^m$ be real-analytic. Define the *solution manifold*

$$\mathcal{S} := \{(\theta, u) \in \Theta \times U \mid F(\theta, u) = 0\}.$$

At points where $\partial_u F$ is invertible, the implicit function theorem gives a local analytic branch $u^*(\theta)$, and we define the closed-loop datum by

$$\Phi^*(\theta) = \nu(\theta, u^*(\theta)), \quad \Lambda^*(\theta) = \Lambda(\theta, u^*(\theta)).$$

A point $(\theta_*, u_*) \in \mathcal{S}$ is a *simple fold* of the projection $\pi: \mathcal{S} \rightarrow \Theta$, $\pi(\theta, u) = \theta$, if $\partial_u F_*$ has rank $m - 1$, with right null vector $e \in \ker \partial_u F_*$ and left null vector $\ell \in \ker \partial_u F_*^\top$, and the fold nondegeneracy conditions

$$(\partial_\theta F_*)^\top \ell \neq 0, \quad \ell^\top \partial_{uu}^2 F_*[e, e] \neq 0$$

hold.

Theorem 6.4 (Local rank-one blow-up near a simple fold). *Let $(\theta_*, u_*) \in \mathcal{S}$ be a simple fold in the sense of Assumption 6.3, and let $(\theta, u^*(\theta)) \in \mathcal{S}$ be a regular branch approaching (θ_*, u_*) from the side where $\partial_u F$ is invertible. Let $\sigma(\theta)$ denote the smallest singular value of $\partial_u F(\theta, u^*(\theta))$, and let e, ℓ be the limiting right and left singular vectors corresponding to $\sigma(\theta) \rightarrow 0$. (Equivalently, along the branch one may choose singular vectors $e(\theta), \ell(\theta)$ for the smallest singular value with $e(\theta) \rightarrow e$ and $\ell(\theta) \rightarrow \ell$; the displayed leading terms are written using their limiting notation.) Then*

$$\partial_\theta u^* = -(\partial_u F)^{-1} \partial_\theta F = -\sigma(\theta)^{-1} e \ell^\top \partial_\theta F + O(1).$$

Consequently,

$$d\Phi^* = \partial_\theta \nu + \partial_u \nu \partial_\theta u^* = -\sigma(\theta)^{-1} (\partial_u \nu e) (\ell^\top \partial_\theta F) + O(1), \quad (11)$$

up to the bounded direct term. If $a := \partial_u \nu_* e \neq 0$, then the closed-loop Fisher metric satisfies

$$\mathcal{G}^* = (d\Phi^*)^\top D^{-1} (d\Phi^*) = \sigma(\theta)^{-2} (a^\top D_*^{-1} a) b b^\top + O(\sigma(\theta)^{-1}), \quad b := (\partial_\theta F_*)^\top \ell.$$

Thus $\mathcal{G}^*$ has one dominant eigenvalue of order σ^{-2} , with dominant eigendirection converging to $b/\|b\|$. If $\partial_u \nu_* e = 0$, the implicit variable still has a rank-one derivative singularity, but the QSD Fisher metric does not diverge at leading order: the σ^{-1} indirect term drops, so $\mathcal{G}^*$ stays bounded at the σ^{-2} and σ^{-1} orders, and any residual dependence is carried solely by the bounded direct term $\partial_\theta \nu$ of Theorem 4.1. The analogous statement for the augmented metric includes the additional rate-channel term, with leading divergence in Λ^* precisely when $\partial_u \Lambda_* e \neq 0$.

Proof. On the regular side $\partial_u F$ is invertible and the implicit function theorem [24, Thm. 3.3.1] gives the analytic branch u^* with $\partial_\theta u^* = -(\partial_u F)^{-1} \partial_\theta F$ from differentiating $F(\theta, u^*) \equiv 0$. Near a simple fold the singular value decomposition of $\partial_u F$ has a single vanishing singular value $\sigma \rightarrow 0$ with right/left singular vectors e, ℓ ; since the fold is simple, the remaining singular values stay bounded away from zero along the branch, so $(\partial_u F)^{-1} = \sigma^{-1} e \ell^\top + O(1)$; the displayed expansion of $\partial_\theta u^*$ follows, with leading coefficient nonzero by the transversality $(\partial_\theta F_*)^\top \ell \neq 0$. Substituting into the chain rule (11) and using the bounded direct term $\partial_\theta \nu$ (Theorem 4.1) gives the stated $d\Phi^*$. If $a := \partial_u \nu_* e \neq 0$, the indirect term is genuinely of size σ^{-1} , and $\mathcal{G}^* = (d\Phi^*)^\top D^{-1} d\Phi^* = \sigma^{-2} (a^\top D_*^{-1} a) b b^\top + O(\sigma^{-1})$ with $b = (\partial_\theta F_*)^\top \ell$, a rank-one σ^{-2} divergence whose dominant eigendirection converges to $b/\|b\|$; if $\partial_u \nu_* e = 0$ the σ^{-1} term drops and the leading divergence is lost. The rate channel is identical with $\partial_u \Lambda_* e$ in place of $\partial_u \nu_* e$. $\square$

Example 6.5 (A closed-form scalar fold). Theorem 6.4 can be exhibited entirely in closed form. Take the scalar self-consistency ($m = 1$) $F(\theta, u) = u^2 - \theta$. For $\theta > 0$ the regular branch is $u^*(\theta) = \sqrt{\theta}$, with $du^*/d\theta = 1/(2\sqrt{\theta})$. The fold is at $(\theta, u) = (0, 0)$, where $\partial_u F = 2u$ vanishes; it is simple in the sense of Assumption 6.3, since $e = \ell = 1$, $\partial_{uu}^2 F = 2 \neq 0$, and $\partial_\theta F = -1 \neq 0$. The smallest singular value is $\sigma(\theta) = |2u^*| = 2\sqrt{\theta} \rightarrow 0$, so $du^*/d\theta = \sigma^{-1}$, the rank-one gain of Theorem 6.4.

Couple this fold to the two-state killed chain of Example 7.1 through its activation rate:

$$Q(u) = \begin{pmatrix} -(a(u) + \kappa) & a(u) \\ b & -b \end{pmatrix}, \quad a(u) = a_0 + \alpha u, \quad F(\theta, u) = u^2 - \theta.$$

Since $\partial_u a = \alpha$, the coupling condition $\partial_u \nu e \neq 0$ of Theorem 6.4 reads $\alpha \partial_a \nu \neq 0$, which holds whenever $\alpha \neq 0$ and the rate genuinely moves the QSD ($\partial_a \nu \neq 0$). The closed-loop QSD is then $\nu^*(\theta) = \nu(a_0 + \alpha\sqrt{\theta})$, so

$$\frac{d\nu^*}{d\theta} = \partial_a \nu \cdot \frac{\alpha}{2\sqrt{\theta}}, \quad \mathcal{G}^*(\theta) = \sum_s \frac{(\partial_\theta \nu_s^*)^2}{\nu_s^*} = \frac{\alpha^2}{4\theta} \sum_s \frac{(\partial_a \nu_s)^2}{\nu_s} + O(\theta^{-1/2}),$$

provided the coefficient $\sum_s (\partial_a \nu_s)^2 / \nu_s$, with ν and $\partial_a \nu$ evaluated at $a = a_0$, is nonzero. Since $\sigma = 2\sqrt{\theta}$, this is exactly the σ^{-2} blow-up of Theorem 6.4, verified by hand on an analytic branch (Figure 1). If instead $\alpha = 0$ (no coupling) then ν^* is constant and $\mathcal{G}^*$ stays bounded through the fold, illustrating the proviso.

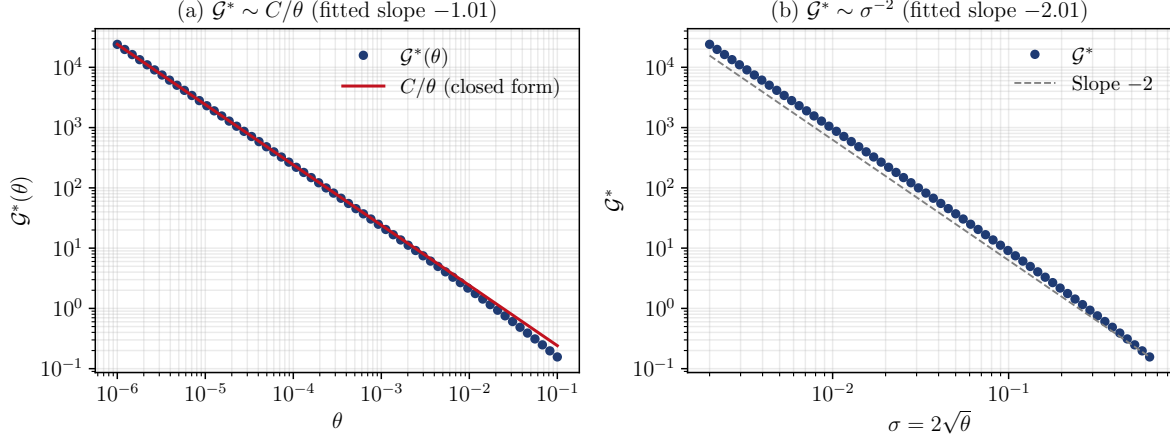


FIGURE 1. The closed-form scalar fold of Example 6.5, verifying Theorem 6.4: the two-state QSD coupled through $a(u) = a_0 + \alpha u$ with $F(\theta, u) = u^2 - \theta$, $u^* = \sqrt{\theta}$, $\sigma = 2\sqrt{\theta}$. (a) The closed-loop Fisher metric $\mathcal{G}^*(\theta)$ tracks the closed-form leading term C/θ with $C = \frac{\alpha^2}{4} \sum_s (\partial_a \nu_s)^2 / \nu_s$ (at $a = a_0$) as $\theta \rightarrow 0$. (b) Equivalently $\mathcal{G}^* \sim \sigma^{-2}$ (fitted slope -2.01 on the asymptotic window $\theta \leq 10^{-3}$, where the leading C/θ dominates the $O(\theta^{-1/2})$ correction). Computed in `figures/scalar_fold.py`.

Remark 6.6 (Stratification interpretation). The previous theorem is local. Globally, if the projection $\pi: \mathcal{S} \rightarrow \Theta$ is proper on the region of interest and satisfies the standard Thom–Boardman genericity assumptions [17], its discriminant $\mathcal{B}$ admits a Whitney stratification [42, 29]: the regular values form open *regimes*, the A_1 fold points form the generic codimension-one walls, and A_2 cusp points and higher singularities form higher-codimension strata. On each regular regime, the closed-loop response geometry is analytic and bounded on compact subsets. Approaching a fold wall along a regular branch, the local behavior is the rank-one blow-up of Theorem 6.4, subject to the observable-coupling condition $\partial_u \nu e \neq 0$ for the QSD metric and $\partial_u \Lambda e \neq 0$ for the rate channel. This is a global interpretation of the local fold theorem under standard genericity hypotheses, not a separate theorem proved here.

Remark 6.7 (Small-noise regularization). When $F = 0$ arises as the deterministic limit of an order-parameter diffusion $du = F(\theta, u) dt + \sigma_{\text{noise}} dW$, the saddle-node normal form along the fold direction fixes the regularized critical-layer width $W \sim \sigma_{\text{noise}}^{2/3}$ and relaxation time $\tau \sim \sigma_{\text{noise}}^{-2/3}$, so the leverage saturates at $\sim \sigma_{\text{noise}}^{-2/3}$ and recovers the divergence of Theorem 6.4 as $\sigma_{\text{noise}} \rightarrow 0$; with $\sigma_{\text{noise}}^2 \propto 1/N$ these read $W \sim N^{-1/3}$, $\tau \sim N^{1/3}$ [41, Ch. X], [26]. These are the standard Fokker–Planck scaling heuristics, recorded for orientation only: nothing in this paper depends on them, and they are not established here at the rigor of §4.

7. EXAMPLES

7.1. The two-state killed chain. Let $T = \{1, 2\}$ and

$$Q = \begin{pmatrix} -(a + \kappa) & a \\ b & -b \end{pmatrix}, \quad a, b > 0, \kappa \geq 0,$$

killing κ from state 1 only, so $q = (\kappa, 0)^\top$ and Q is irreducible iff $a, b > 0$, properly killed iff $\kappa > 0$.

Example 7.1 (Closed-form principal datum and response). The characteristic polynomial gives eigenvalues $\lambda_\pm = \frac{1}{2}[-(a + \kappa + b) \pm \sqrt{(a + \kappa - b)^2 + 4ab}]$, and the principal datum is $-\Lambda = \lambda_+$:

$$\Lambda = \frac{1}{2}[(a + \kappa + b) - \sqrt{(a + \kappa - b)^2 + 4ab}] > 0.$$

The left/right Perron vectors (normalized $\nu \mathbf{1} = 1$, $\nu h = 1$) are

$$\nu \propto (b, a + \kappa - \Lambda), \quad h \propto \left(1, \frac{b}{b - \Lambda}\right),$$

both strictly positive since $a + \kappa - \Lambda = \frac{1}{2}[(a + \kappa - b) + \sqrt{(a + \kappa - b)^2 + 4ab}] > 0$ and $b - \Lambda = \frac{1}{2}[-(a + \kappa - b) + \sqrt{(a + \kappa - b)^2 + 4ab}] > 0$; the two equivalent forms $h_2/h_1 = (a + \kappa - \Lambda)/a = b/(b - \Lambda)$ coincide by the characteristic identity $\Lambda^2 - (a + \kappa + b)\Lambda + \kappa b = 0$. The Hellmann–Feynman and reduced-resolvent formulas (4)–(5) reproduce $\partial\Lambda/\partial a$, $\partial\Lambda/\partial\kappa$ and $\partial\nu/\partial a$ exactly; we verified this symbolically and numerically. As $\kappa \rightarrow 0$: $\Lambda \rightarrow 0$, $h \rightarrow \mathbf{1}$, $\nu \rightarrow \pi = (b, a)/(a + b)$, $R \rightarrow Q_0^\#$, and (5) becomes the conservative $\partial_a \pi_1 = -b/(a + b)^2$ of Remark 4.7, with the positive sign $+\pi Q_a Z$.

7.2. The scaling gauge and the channel split. The two response coordinates ν and Λ have genuinely different kernels (Figure 2). The cleanest witness is the overall time-scaling symmetry, present in every family.

Proposition 7.2 (Channel split). *Let $s > 0$ act on a sub-generator by $Q \mapsto sQ$. Then the principal datum transforms as*

$$\nu(sQ) = \nu(Q), \quad h(sQ) = h(Q), \quad \Lambda(sQ) = s\Lambda(Q), \quad R(sQ) = s^{-1}R(Q).$$

Consequently, for any family containing a scaling direction $\partial_{\log s}$ at θ ,

$$\partial_{\log s} \nu = 0 \quad \text{but} \quad \partial_{\log s} \Lambda = \Lambda \neq 0.$$

Hence $\partial_{\log s} \in \ker J = \ker \mathcal{G}$ but $\partial_{\log s} \notin \ker \nabla \Lambda$: the distributional and rate channels of Remark 5.6 have distinct kernels, and the augmented metric $\mathcal{G}^\oplus$ of (10) is strictly larger than $\mathcal{G}$ in the scaling direction.

Proof. sQ has eigenvalues $s \operatorname{spec}(Q)$ with identical eigenvectors, so $\Lambda(sQ) = s\Lambda(Q)$ and ν, h are unchanged; $R(sQ) = (sQ + s\Lambda I)^\# = (s(Q + \Lambda I))^\# = s^{-1}(Q + \Lambda I)^\# = s^{-1}R$ by the scaling law of the group inverse. Differentiating in $\log s$ at $s = 1$ gives the displayed derivatives, whence the kernel statements; $\mathcal{G}^\oplus - \mathcal{G} = w d(\log \Lambda)^{\otimes 2}$ evaluates to $w \neq 0$ on $\partial_{\log s}$ since $\partial_{\log s} \log \Lambda = 1$. $\square$

Remark 7.3. Proposition 7.2 is the structural reason the decay rate is worth carrying as a separate observable: it sees the overall time scale, to which the conditional law ν is blind. More generally, any one-parameter group of *rate symmetries* of the family (gauge directions that rescale the generator without changing its conditional law) lies in $\ker J$ but not in $\ker \nabla \Lambda$, and produces an exact non-identifiability leaf (Theorem 6.1) for the distributional channel.

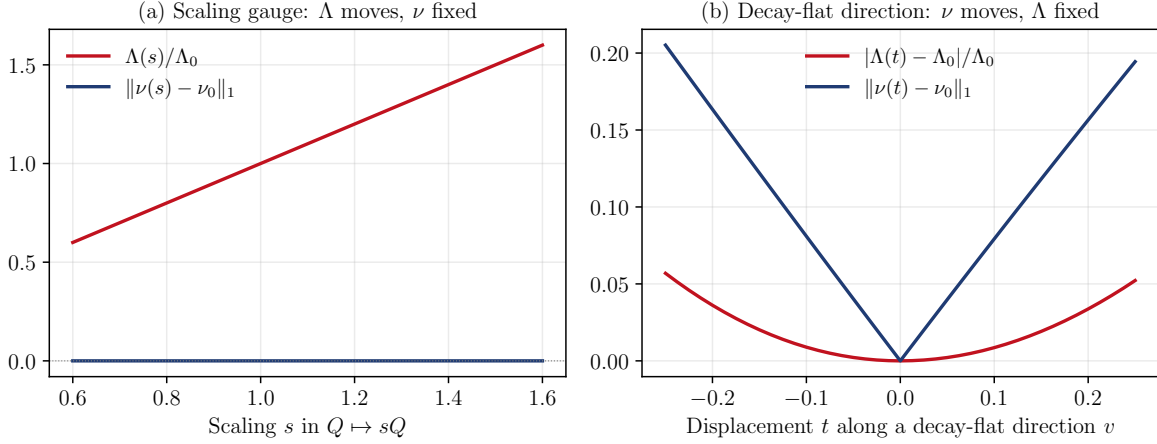


FIGURE 2. The QSD and decay-rate channels are distinct (Definition 5.5, Proposition 7.2), on the two-state killed chain $Q(a, b, \kappa)$. **(a)** Along the scaling gauge $Q \mapsto sQ$ the quasi-stationary distribution ν is exactly invariant ($\|\nu(s) - \nu_0\|_1 = 0$) while the decay rate scales linearly — a perturbation in $\ker J_\nu$ but not $\ker \nabla \Lambda$. **(b)** Along a direction in $\ker \nabla \Lambda$, the decay rate is stationary to first order while ν moves — a perturbation in $\ker \nabla \Lambda$ but not $\ker J_\nu$. A perturbation may thus move Λ without moving ν , or conversely. Computed in `figures/channel_split.py`.

7.3. A self-consistent three-state fold.

Example 7.4 (Fold-induced divergence). Let $T = \{1, 2, 3\}$ with a killed cycle whose activation rate is modulated by an order parameter $u \in (0, 1)$ through a sigmoidal feedback, e.g.

$$Q(\theta, u) = \begin{pmatrix} -(\alpha(\theta, u) + \kappa_1) & \alpha(\theta, u) & 0 \\ \beta & -(\beta + \gamma) & \gamma \\ \delta & 0 & -\delta \end{pmatrix}, \quad \alpha(\theta, u) = \alpha_0 e^{g\varsigma(u-\theta)},$$

with ς a smooth increasing saturating (sigmoidal) function and a single small killing $\kappa_1 > 0$. Couple u self-consistently to the activation prevalence by $F(\theta, u) = \nu_1(\theta, u) - u$ (or any analytic closure with $\partial_u F$ changing sign). For a range of the gain g the equation $F = 0$ has the S-shaped graph of a fold: two stable branches separated by a fold locus $\mathcal{B} = \{\partial_u F = 0\}$ in θ . On each open regime $\Theta \setminus \mathcal{B}$, Theorem 6.4 gives bounded analytic closed-loop response; approaching $\mathcal{B}$, $\partial_\theta u^*$ diverges rank-one, and — since $\partial_u \nu$ couples to the fold here — $d\Phi^*$, $\nabla \Lambda^*$, $\mathcal{G}^*$ inherit it along the fold direction, with the dominant eigenvector of $\mathcal{G}^*$ aligning with $v_* \propto (\partial_\theta F)^\top \ell$. The univariate ($m = 1$) case makes $\ell = e = 1$ and $\sigma = |\partial_u F|$, so $\partial_\theta u^* = -\partial_\theta F / \partial_u F$ and the divergence is the textbook implicit-function blow-up at a fold, here transported through ν and Λ to the full information geometry. This is the local model behind Remark 6.6: parameter space splits into regimes of qualitatively distinct quasi-stationary behavior, separated by the fold wall on which the response is maximally sensitive.

A verified instance. Take $\varsigma = \tanh$ and the concrete parameters

$$(\beta, \gamma, \delta, \kappa_1, \alpha_0, g) = (1, 0.8, 1.2, 0.05, 1, -5).$$

Since ν_1 depends on (θ, u) only through $w := u - \theta$, the equilibrium branch is traced parametrically: with $G(w) := \nu_1(\alpha_0 e^{g\varsigma(w)})$ it is $\{(\theta, u) = (G(w) - w, G(w))\}$, and the fold condition $\partial_u F = 0$ reads $G'(w) = 1$. Here $\max_w G'(w) \approx 1.24 > 1$, so two folds occur (at $w \approx \pm 0.18$) and the branch

is bistable (Figure 3(a)). Approaching a fold along the branch, a direct computation from (5), (4), and the group inverse $R = (A + P)^{-1}(I - P)$ confirms the predicted scalings: log-log regression of the response norms against $\sigma = |\partial_u F|$ over three decades ($\sigma \in [2.3 \times 10^{-4}, 2.4 \times 10^{-1}]$) gives slopes -1.03 , -1.03 , and -2.03 for $\|\mathrm{d}\Phi^*\|$, $|\nabla\Lambda^*|$, and $\lambda_{\max}(\mathcal{G}^*)$ respectively, matching the predicted $-1, -1, -2$ of Theorem 6.4 (Figure 3(b)). The computation uses only the response calculus of §4 and is reproducible via the script `figures/fold_example.py`.

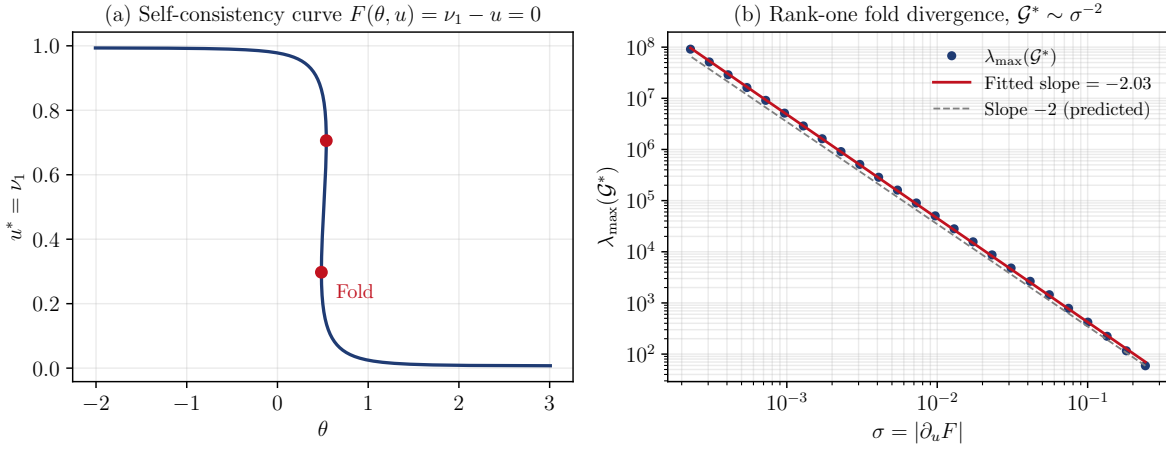


FIGURE 3. Worked instance of the self-consistent three-state fold (Example 7.4), with $\varsigma = \tanh$ and $(\beta, \gamma, \delta, \kappa_1, \alpha_0, g) = (1, 0.8, 1.2, 0.05, 1, -5)$. (a) The self-consistency curve $F(\theta, u) = \nu_1(\theta, u) - u = 0$ is S-shaped; the two red dots are the folds $\partial_u F = 0$ bounding the bistable window. (b) Approaching a fold, the closed-loop Fisher metric diverges as $\mathcal{G}^* \sim \sigma^{-2}$ in $\sigma = |\partial_u F|$ (fitted slope -2.03 against the predicted -2), the rank-one divergence of Theorem 6.4; the norms $\|\mathrm{d}\Phi^*\|$ and $|\nabla\Lambda^*|$ scale as σ^{-1} (slopes -1.03). Computed solely from the response formulas of §4.

8. INFINITE-DIMENSIONAL GENERATORS

Status of this section. The statements are split by proof obligation. Proposition 8.2 is a *cited assembly*: under the standing hypotheses of Assumption 8.1, cited operator theory (Champagnat–Villemonais, Krein–Rutman, Meyn–Tweedie) yields the isolated simple principal eigentriple, the bounded reduced resolvent, and the gap-as-conditional-ergodicity-rate identification. Proposition 8.3 (the operator-valued derivative formulas) and Theorem 8.4 (fold amplification and the $\Lambda \rightarrow 0$ recovery) are then *conditional transport*: elementary deductions from an isolated simple eigentriple and a bounded reduced resolvent, borrowing no further QSD theory — Theorem 8.4’s recovery carrying the additional gap-persistence hypothesis stated there. Assumption 8.1 is discharged concretely on the exactly-solvable countable instance Example 8.6. This is an *assembly* of cited theorems for the countable case, not a new theorem. Remark 8.5 (the diffusion case) is a roadmap at the rigor of a sketch and is *not* part of the proved package.

Everything above is finite-dimensional, where Perron–Frobenius supplies simplicity and the gap for free. The natural objection is that the finite Metzler setting switches off every analytic difficulty — non-compactness, unboundedness, the principal eigenvalue merging into the essential spectrum — that makes infinite-dimensional QSD theory hard. This section lifts the response geometry to non-compact state spaces, the missing ingredients supplied by a Foster–Lyapunov

drift in place of finite Perron–Frobenius; as the status note records, the countable case is a rigorous corollary discharged on Example 8.6, the diffusion case a sketch (Remark 8.5). Companion proof notes, available with the source, hold the routine verifications.

8.1. Setting and hypotheses. Let E be a Polish state space (countable discrete, or a domain in $\mathbb{R}^d$), ∂ an absorbing cemetery, and (X_t) a μ -irreducible Markov process on $E \cup \{\partial\}$ with sub-Markovian generator L on E (an infinite Metzler matrix in the discrete case; a second-order elliptic operator with killing potential $\kappa \geq 0$ in the diffusion case). The new feature is unboundedness: L is unbounded, E is infinite, and there is no a priori spectral gap. The control object is a *Lyapunov (Foster) weight* $W: E \rightarrow [1, \infty)$ with $W \rightarrow \infty$ at infinity, in whose weighted space $B_W = \{f : \|f\|_W := \sup_x |f(x)|/W(x) < \infty\}$ the spectral theory is recovered.

Assumption 8.1 (Standing hypotheses). (H1) P_t is irreducible and positivity-improving. (H2) The killing is accessible and not identically zero. (H3) *Foster–Lyapunov drift*: there exist W as above, constants $c > 0$, $b < \infty$, and a small set $K \subset E$ with $LW \leq -cW + b\mathbf{1}_K$ on E . (H4) *Doebelin minorization on K* : there are $t_0 > 0$, $\theta > 0$, and a probability ν_0 with $P_{t_0}(x, \cdot) \geq \theta\nu_0(\cdot)$ for all $x \in K$.

(H3)–(H4) are exactly the Champagnat–Villemonais conditions; (H1)–(H2) supply irreducibility and a genuine killed process. The compact case is $W \equiv 1$, $K = E$; the finite-state case of §2 is E finite, $W \equiv 1$.

Proposition 8.2 (QSD existence, simple isolated eigentriple, bounded reduced resolvent — cited corollary). *This is an assembly of cited QSD theory, not a new deduction. Under Assumption 8.1 there is a unique quasi-stationary distribution ν on E and a principal eigentriple (Λ, ν, h) with $\Lambda > 0$, $\nu L = -\Lambda\nu$, $Lh = -\Lambda h$, $h \in B_W$, $h > 0$, $\nu(h) = 1$, such that $-\Lambda$ is a real, algebraically simple, isolated eigenvalue of L on B_W with the rest of the spectrum in $\{\Re z \leq -\Lambda - g\}$ for a gap $g > 0$; consequently the reduced resolvent $R = (L + \Lambda I)^\#$ is bounded on B_W . The gap is the conditional-ergodicity rate: for every μ with $\mu(W) < \infty$, $\|P_\mu(X_t \in \cdot \mid t < \tau_\partial) - \nu\|_{\text{TV}} \leq C(\mu)e^{-gt}$ with the same g . The one genuinely nontrivial borrowed step is the passage from the Foster drift (H3) to a spectral gap for the sub-Markovian (killed, not conservative) semigroup P_t on B_W : for the killed process this gap — equivalently the exponential conditional ergodicity below, together with existence and uniqueness of ν — is supplied by the Champagnat–Villemonais criterion [8, 9] (the general weighted-space Foster–Lyapunov drift framework being Meyn–Tweedie [32], which is conservative), and the simple strictly positive principal eigenfunction is Krein–Rutman [25].*

Proposition 8.3 (Operator-valued response calculus — conditional transport). *Assume an isolated, algebraically simple principal eigentriple (Λ, ν, h) with a bounded reduced resolvent $R = (L + \Lambda I)^\#$ on B_W (as supplied by Proposition 8.2), and that $L = L(p)$ is a holomorphic family of type (A) in a control p (e.g. $\partial_p L$ relatively L -bounded). Then (Λ, ν, h) are real-analytic in p and*

$$\partial_p \Lambda = -\nu(\partial_p L)h, \quad \partial_p \nu = -\nu(\partial_p L)R + c_p \nu, \quad c_p = \nu(\partial_p L)R\mathbf{1},$$

identical in form to the finite-state formulas of Theorem 4.1, with R the group inverse of the unbounded generator on B_W . This is an elementary deduction from the displayed hypotheses, conditional on them; it borrows no further QSD theory.

Theorem 8.4 (Fold amplification and the $\Lambda \rightarrow 0$ recovery). *On a stable self-consistent branch with slow eigenvalue $\sigma \rightarrow 0$, the finite-state amplification holds verbatim: $\partial_p u_* = F_p/\sigma$, $\partial_p \Lambda = \Lambda'(u_*)F_p/\sigma + O(1)$, and the pulled-back Fisher metric is $\Theta(\sigma^{-2})$, with the same observable-coupling condition. Moreover, if the conservative generator L_0 (killing scaled to zero) is itself exponentially ergodic in B_W and, in addition, its weighted-space spectral gap persists under the killing-scaling perturbation, i.e. $\liminf_{\Lambda \rightarrow 0} g(\Lambda) \geq g_0 > 0$, then as $\kappa \downarrow 0$ one has $\Lambda \rightarrow 0$, $\nu \rightarrow \pi$, and $R \rightarrow (L_0)^\#$*

in operator norm: the killed response calculus converges to the stationary (Schweitzer–Meyer) calculus, the infinite-dimensional analogue of Theorem 4.6. (The gap-persistence hypothesis is automatic in finite dimensions; in B_W it is an assumption, not a conclusion — see the proof.)

Proof. Proposition 8.2 is the cited assembly under Assumption 8.1; Proposition 8.3 and Theorem 8.4 are then elementary deductions *conditional on its conclusions* (an isolated simple eigentriple and a bounded reduced resolvent). We give all three here (companion proof notes hold only the routine verifications). For the countable case the hypotheses (H1)–(H4) are moreover discharged concretely in Example 8.6, so the assembled eigentriple is realized, not merely conditional.

Proposition 8.2. (H3)–(H4) are exactly the Champagnat–Villemonais hypotheses [8, 9], which give existence and uniqueness of the quasi-stationary distribution ν and exponential conditional ergodicity $\|P_\mu(X_t \in \cdot \mid t < \tau_\partial) - \nu\|_{\text{TV}} \leq C(\mu)e^{-gt}$ for every μ with $\mu(W) < \infty$ — the killed-process spectral gap $g > 0$ on B_W . (Mechanistically the Foster drift (H3), $LW \leq -cW + b\mathbf{1}_K$, is what compactifies the killed semigroup $P_t: B_W \rightarrow B_W$: outside the small set K the drift contracts W , so P_t maps the unit ball of B_W into a relatively compact set up to an arbitrarily small tail; the general weighted-space Foster–Lyapunov drift framework is Meyn–Tweedie [32] in the conservative case, and the sub-Markovian gap is the Champagnat–Villemonais content cited above.) By (H1)–(H2) the resolvent is positivity-improving on the cone of W -dominated nonnegative functions, so Krein–Rutman [25] yields a simple, strictly positive principal eigenfunction $h \in B_W$, $h > 0$, with eigenvalue $-\Lambda$ ($\Lambda > 0$ because the killing is accessible and nonzero, (H2)) that is isolated from the rest of the spectrum by a gap $g > 0$ equal to the conditional-ergodicity rate. Normalizing $\nu L = -\Lambda\nu$, $Lh = -\Lambda h$, $\nu(h) = 1$ gives the principal eigentriple.

Proposition 8.3. Isolation and simplicity make the Riesz projection $\Pi = h \otimes \nu$ rank-one and, since $\partial_p L$ is relatively L -bounded, $L(p)$ is a holomorphic family of type (A) on B_W ; Kato’s theory of a simple isolated eigenvalue [22] then makes (Λ, ν, h) and Π real-analytic in p . The reduced resolvent $R = (L + \Lambda I)^\#$ is bounded on B_W precisely because of the gap $g > 0$. Differentiating $Lh = -\Lambda h$ and $\nu L = -\Lambda\nu$ and projecting off Π with R reproduces, verbatim, the finite-state computation of Theorem 4.1: $\partial_p \Lambda = -\nu(\partial_p L)h$ and $\partial_p \nu = -\nu(\partial_p L)R + c_p \nu$, $c_p = \nu(\partial_p L)R\mathbf{1}$, all finite because R is bounded.

Theorem 8.4. The amplification argument of §6 used only the scalar implicit function theorem applied to the self-consistency relation and the finiteness of $\partial_p \Lambda, \partial_p \nu$; both are now available from Proposition 8.3, so $\partial_p u_* = F_p/\sigma$, $\partial_p \Lambda = \Lambda'(u_*)F_p/\sigma + O(1)$, and the pulled-back Fisher metric is $\Theta(\sigma^{-2})$, with the same observable-coupling condition. For the $\Lambda \rightarrow 0$ recovery, scaling the killing $\kappa \downarrow 0$ is an analytic perturbation of the isolated simple eigenvalue 0 of the conservative generator L_0 . We *assume* the weighted-space spectral gap persists under this perturbation, $\liminf_{\Lambda \rightarrow 0} g(\Lambda) \geq g_0 > 0$ (the stated hypothesis): in finite dimensions this is automatic by continuity of the simple isolated eigenvalue, but in B_W the killing perturbation is not relatively compact in general, so persistence of the gap is an assumption rather than a consequence of ergodicity of L_0 alone. Under it, $\Lambda \rightarrow 0$, $\nu \rightarrow \pi$, $R \rightarrow (L_0)^\#$ in operator norm, giving the stationary (Schweitzer–Meyer) calculus, the infinite-dimensional analogue of Theorem 4.6.

Status. For the countable case (H1)–(H4) are *verified*, not merely assumed, on Example 8.6, so the corollary is realized; the diffusion case is the sketch of Remark 8.5, and the further fold-sensing residual (the functional-CLT step, companion paper [35]) is Remark 8.7. $\square$

Remark 8.5 (Diffusion extension — roadmap, not a proved result). Let $E \subseteq \mathbb{R}^d$ be a domain and $L = \frac{1}{2}\nabla \cdot (a\nabla \cdot) + b \cdot \nabla - \kappa$ a uniformly elliptic second-order operator with killing potential $\kappa \geq 0$, $\kappa \not\equiv 0$, on a weighted space B_W with $W \rightarrow \infty$ at ∂E and at infinity. Suppose $\partial_p L$ is relatively L -bounded on B_W and the Foster drift (H3) and minorization (H4) hold for the killed diffusion. Then the conclusions of Propositions 8.2, 8.3 and Theorem 8.4 are *expected* to hold for L , but we

do not establish them at the rigor of §§2–6: the open inputs are (i) the verification of (H3)–(H4) for the specific elliptic L (standard for confining drifts, e.g. Ornstein–Uhlenbeck with killing, but model-dependent), and (ii) the control of the essential spectrum showing the principal eigenvalue is isolated in B_W and the group inverse bounded there. Both are classical (Champagnat–Villemonais for (i); Agmon/weighted-space spectral theory for (ii)) but are operator-theoretic verifications we record only in sketch (in the companion proof notes); the countable case (Propositions 8.2, 8.3 and Theorem 8.4, Example 8.6) is the one carried out in full. We state this as a roadmap, not a proposition, precisely because (i)–(ii) are not discharged here.

8.2. An explicit exactly-solvable instance. The hard infinite-dimensional ingredient is the Foster–Lyapunov drift (H3) on a genuinely non-compact space. We exhibit it explicitly.

Example 8.6 (Birth–death–with-killing chain). Take the active-count $n \in \{0, 1, 2, \dots\}$ as a continuous-time chain with birth $n \rightarrow n + 1$ at constant rate β , death $n \rightarrow n - 1$ at rate μn , and killing $n \rightarrow \partial$ at rate κn (absorbing). The living generator is $(Lf)(n) = \beta(f(n+1) - f(n)) + \mu n(f(n-1) - f(n)) - \kappa n f(n)$; each living row sums to $-\kappa n$, so the QSD identity $\Lambda = -\nu L \mathbf{1} = \kappa \mathbb{E}_\nu[N]$ holds exactly. For the weight $W(n) = e^{\gamma n}$,

$$\frac{(LW)(n)}{W(n)} = \beta(e^\gamma - 1) - n[\mu(1 - e^{-\gamma}) + \kappa],$$

affine in n with strictly negative slope for every $\gamma > 0$, so (H3) holds with explicit $N_0 = \lceil (\beta(e^\gamma - 1) + c)/(\mu(1 - e^{-\gamma}) + \kappa) \rceil$, $K = \{0, \dots, N_0 - 1\}$, and $b = \max_{n \in K} (D(n) + c)W(n)$. The remaining hypotheses are immediate: the chain is irreducible and positivity-improving on $\{0, 1, 2, \dots\}$ (birth and death connect adjacent states, $\beta, \mu > 0$), giving (H1); the killing κn is accessible and not identically zero for $\kappa > 0$, giving (H2); and (H4) is the Doeblin minorization on the *finite* set K , automatic for an irreducible chain. Thus all of (H1)–(H4) are discharged *concretely*, not assumed. The recovery rate μn grows without bound, so the chain “comes down from infinity” and Proposition 8.2 applies on this genuinely non-compact space. Because both down-rates are linear in n , the instance is *exactly solvable*: with $\rho := \beta/(\mu + \kappa)$,

$$\nu = \text{Poisson}(\rho), \quad \Lambda = \frac{\kappa\beta}{\mu + \kappa} = \kappa\rho, \quad \text{spectral gap} = \mu + \kappa,$$

since the choice $\rho = \beta/(\mu + \kappa)$ makes the m -coefficient of the per-state eigen-relation vanish (this is why Poisson is stationary) and the constant equals $-\Lambda$. For the calibrated instance $(\beta, \mu, \kappa, \gamma, c) = (1, \frac{1}{2}, \frac{1}{10}, 1, 1)$ one gets $N_0 = 7$, $b \approx 94.7$, $\rho = \frac{5}{3}$, $\Lambda = \frac{1}{6}$, $\text{gap} = \frac{3}{5}$, and exponential moment $\sum_n \nu(n)e^{\gamma n} = e^{\rho(e-1)} \approx 17.53$; a deterministic truncation reproduces Λ and the gap to 10^{-15} and the identity $\Lambda = \kappa \mathbb{E}_\nu[N]$ to 5×10^{-15} (Figure 4). The rational drift threshold $\bar{D}(7) = -91/80 \leq -1$ and the exact (Λ, gap) /eigen-identities are rational and reproducible in `UnboundedLyapunovKernel.lean`.

Remark 8.7 (Scope of §8). Two residuals lie beyond the countable assembly (whose status is the note opening this section; the infinite-dimensional content over the finite case is that quasi-compactness, isolation, and the bounded group inverse come from the Lyapunov drift rather than for free). The *diffusion* case (Remark 8.5) is at the rigor of a sketch, its open inputs the operator-theoretic verification of (H1)–(H4) and the essential-spectrum control for elliptic L . The infinite-dimensional *fold-sensing* limit (companion paper [35]) carries a second: replacing the finite- N demographic fluctuation by the functional-CLT fluctuation of the empirical occupation measure is quantitative ($O(N^{-1/2})$) in the generic regime $\xi \gg 1$, but on the band $\xi = O(1)$ only a numerically checked generator-convergence with a $\sim N^{-2/3}$ error. The finite-truncation algebraic kernel — the group-inverse identity $(L + \Lambda I)R = I - h \otimes \nu$ and the response formulas

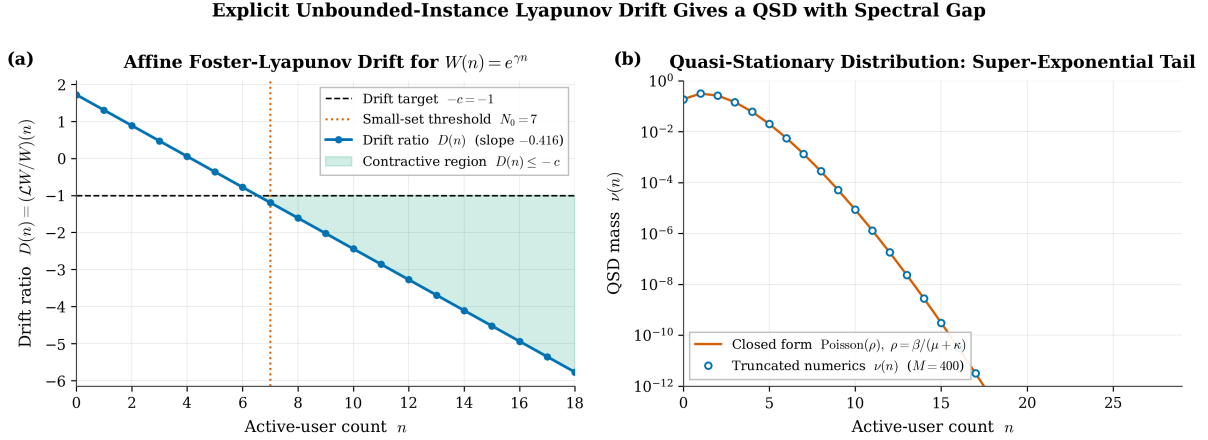


FIGURE 4. The explicit non-compact Foster–Lyapunov instance of Example 8.6 (calibrated $(\beta, \mu, \kappa, \gamma) = (1, \frac{1}{2}, \frac{1}{10}, 1)$). **(left)** The drift ratio $D(n) = (LW/W)(n)$ for $W(n) = e^{\gamma n}$ is affine in n with strictly negative slope, crossing the level $-c = -1$ at $N_0 = 7$; the shaded region marks $\{n \geq N_0\}$ where $D(n) \leq -c$, certifying $LW \leq -cW + b\mathbf{1}_K$ with the finite small set $K = \{0, \dots, 6\}$. **(right)** The resulting quasi-stationary distribution $\nu(n)$ (log scale) is Poisson(ρ) with $\rho = \frac{5}{3}$, exhibiting the super-exponential tail that confines the chain on the unbounded state space. The exact $\Lambda = \frac{1}{6}$ and gap $\frac{3}{5}$ cross-check Proposition 8.2 in closed form. Computed in `analysis/unbounded_qsd_lyapunov.py`; rational core reproducible in `UnboundedLyapunovKernel.lean`.

— is reproducible in `InfiniteDimQSDKernel.lean` (exact on the two-state instance; residuals $\leq 2 \times 10^{-10}$ on the model’s 5×5 killed generator).

9. CONCLUDING REMARKS

9.1. Summary. For a real-analytic family of irreducible Metzler sub-generators we have shown that the principal spectral datum (Λ, ν) — decay rate and quasi-stationary distribution — is real-analytic on the irreducible locus, that its simplicity is rigid (Proposition 2.6), and that its first-order response is carried by the spectral projection and reduced resolvent (Theorem 4.1). The QSD map endows parameter space with a canonical Fisher information metric, equal to the pulled-back Fisher–Rao metric and the KL Hessian, whose kernel is the non-identifiability foliation (Theorems 5.2, 6.1); adjoining the decay rate adds a rank-one rate channel (Definition 5.5). Under a self-consistent coupling the response acquires a rank-one singularity at simple folds, transmitted to the QSD channel when $\partial_u \nu e \neq 0$ and to the decay-rate channel when $\partial_u \Lambda e \neq 0$, and under genericity the discriminant carries a stratified-regime interpretation (Theorem 6.4 and Remark 6.6). The classical group-inverse sensitivity theory of stationary distributions is the no-killing boundary case (Theorem 4.6). Beyond the finite, noise-free core, the entire calculus lifts to non-compact countable and diffusion state spaces, with an explicit exactly-solvable Foster–Lyapunov instance (Propositions 8.2, 8.3 and Theorem 8.4, Example 8.6). The companion paper [35] carries the deterministic fold blow-up into the finite- N regime, where it is regularized by an exactly self-similar within-basin width $W_N = (D/c)^{1/3} G(\xi)$, and shows that the σ^{-2} amplification and the finite- N width are one object — a fold-sensing resolution floor $R(\xi) = 2G(\xi)/\xi$, a Cramér–Rao bound on distance-to-fold estimation with a $\Theta(N^{-1/3})$ floor.

9.2. The motivating application. The framework was distilled from a problem in mathematical epidemiology: a finite-state model of a metastable behavioral population in which one state leaks to an absorbing “death” state, so the long-run object is a quasi-stationary distribution and a mortality-like decay rate. There, ν is the conditional prevalence among survivors, Λ the per-capita hazard, the parameters θ are intervention-controllable rates, the information metric $\mathcal{G}$ is an *intervention-efficiency* tensor, its kernel encodes interventions with no distributional effect, the self-consistent coupling is a mean-field prevalence feedback whose fold is a tipping point, and the rank-one divergence is the outsized leverage of interventions delivered near criticality. The *channel split* (Proposition 7.2) there expresses that some interventions move mortality without moving the conditional case-mix and conversely — e.g. a harm-reduction measure that lowers Λ with little effect on ν versus an enforcement measure transverse to both. We have deliberately stated the theory with no such interpretation attached: it applies to any parametrized family of killed finite Markov generators — metastable chemical kinetics, reliability and absorbing queueing models, population genetics with extinction, evolutionary dynamics with absorbing fixation — wherever the quasi-stationary datum is the observable and its parameter dependence is the question.

9.3. Extensions. The infinite-state extension once listed here as future work is now carried out in §8: for sub-Markovian generators on countable or continuous state spaces with a Foster–Lyapunov drift, Kato’s theory and the reduced-resolvent calculus carry over to the operator setting, the rigidity supplied by Krein–Rutman and quasi-compactness in place of finite Perron–Frobenius, with an explicit exactly-solvable instance (Example 8.6). Two further generalizations remain. (1) *Higher-order information geometry*: the α -connections and the dual-flat structure of the response model invite a study of the embedding curvature of $\Phi(\Theta_{\text{reg}})$ in Δ° , i.e. how far the QSD family is from an exponential family, and of the m -projection estimator of Remark 5.7 as a statistical procedure. (2) *Cusp and higher catastrophes*: Remark 6.6 isolates the fold as the generic wall, but cusps (A_2) and swallowtails (A_3) appear in codimension and carry their own normal-form scalings for the response; classifying the information-geometric signature of each elementary catastrophe is a natural sequel.

Code availability. The source code (figure and verification scripts, the walkthrough notebook, and the Lean 4 kernels) is available at the project [REPOSITORY](#); the kernels and notebook are also bundled as arXiv ancillary files. All numerical claims in the paper are reproducible from the ancillary code: `verify_identities.py` checks every displayed first- and second-order identity against central finite differences on randomly generated irreducible Metzler families (Remark 4.8); `figures/fold_example.py` regenerates the worked fold of Example 7.4 (Figure 3) and refuses to emit the figure unless the fitted exponents match the predicted $(-1, -1, -2)$ within tolerance; and a fully executed companion notebook (`qsd_response_geometry_walkthrough.ipynb`) restates each principal result under its printed number and re-verifies it numerically. Every displayed first- and second-order identity, including the right-eigenvector response (6), is checked in both the script and the notebook; `analysis/eflat_nonuniform_qsd.py` verifies the e -flat non-uniform construction of Example 5.10 (exponential-family QSD, $h \neq \mathbf{1}$, active c_j , and $\mathcal{G} = \text{Cov}$) at 2×10^3 random points. No external data are used. The infinite-dimensional extension of §8 is reproduced by `analysis/unbounded_qsd_lyapunov.py` and `analysis/infinite_dim_qsd_kernel.py` (the non-compact instance and the algebraic kernel, Figure 4); the load-bearing rational identities are machine-checked in the Lean 4 [43] kernels `UnboundedLyapunovKernel.lean` and `InfiniteDimQSDKernel.lean`, which ship self-contained in the companion subfolder `lean/` with an executable walkthrough in `notebook/`. The Lean layer verifies only the *rational core* over $\mathbb{Q}$ —the group-inverse identity $(Q + \Lambda I)R = I - h \otimes \nu$ on the finite instance and Example 8.6’s exactly-solvable Poisson QSD ($\Lambda = 1/6$, gap $3/5$, $N_0 = 7$)—and *not* the analytic Krein–Rutman/Kato

content on the weighted space, the infinite-dimensional limit, or Champagnat–Villemonais ergodicity, which are cited; the per-kernel scope is in `lean/MAPPING.md` and the axiom profile in `lean/AXIOMS.md`. The finite- N and fold-sensing analysis (`at_fold*.py`, `fold_sensing*.py`), the FLINT/Arb interval certificates (continuum gap, variance band, and the new *pending* discrete-gap certificate `tools/cert_discrete_gap.py`), and the Agmon/Kato/discrete-Sturm rational kernels ship with the companion paper [35].

APPENDIX A. THE GROUP INVERSE: EXISTENCE, UNIQUENESS, AND THE RESOLVENT INTEGRAL

For completeness we record the facts about the group inverse used above; see [10, Ch. 4], [31] for proofs.

Lemma A.1. *Let $A \in \mathbb{R}^{n \times n}$ have 0 as a semisimple (index-1) eigenvalue, so $\mathbb{R}^n = \ker A \oplus \operatorname{Im} A$. Then there is a unique matrix $A^\#$ with $AA^\# = A^\#A$, $A^\#AA^\# = A^\#$, $AA^\#A = A$; it acts as $(A|_{\operatorname{Im} A})^{-1}$ on $\operatorname{Im} A$ and as 0 on $\ker A$. If P_0 is the spectral projection onto $\ker A$ along $\operatorname{Im} A$, then $A^\# = (A + P_0)^{-1}(I - P_0)$ and $AA^\# = I - P_0$. When $A = Q + \Lambda I$ for $Q \in \mathcal{M}_{\text{irr}}(T)$ with principal eigenvalue $-\Lambda$, the hypotheses hold with $P_0 = P = h\nu$, giving the reduced resolvent $R = A^\#$ of Definition 3.2, and the contour representation $R = \frac{1}{2\pi i} \oint_\gamma \frac{(zI - Q)^{-1}}{z + \Lambda} dz$, γ a small circle about $-\Lambda$ enclosing no other eigenvalue.*

Proof. Existence/uniqueness and the projection formula are standard for index-1 matrices [10, Thm. 4.5.2]. For the contour formula, the resolvent has the Laurent expansion $(zI - Q)^{-1} = \frac{P}{z + \Lambda} + H(z)$ near $z = -\Lambda$, where P is the residue (the spectral projection) and H is holomorphic at $-\Lambda$ with $H(-\Lambda) = R$ the reduced resolvent (this is the defining property of R in Kato's Laurent expansion, no nilpotent part since the eigenvalue is simple). Therefore $\frac{(zI - Q)^{-1}}{z + \Lambda} = \frac{P}{(z + \Lambda)^2} + \frac{H(z)}{z + \Lambda}$, whose residue at $z = -\Lambda$ is $0 + H(-\Lambda) = R$; the contour integral equals this residue by the residue theorem. $\square$

APPENDIX B. KATO ANALYTICITY FOR A SIMPLE ISOLATED EIGENVALUE

Lemma B.1 ([22, Ch. II §1.8, Ch. VII §1.3]). *Let $\theta \mapsto A(\theta) \in \mathbb{C}^{n \times n}$ be real-analytic on an open set Ω , and suppose $A(\theta_0)$ has an algebraically simple eigenvalue λ_0 isolated from the rest of the spectrum by a Jordan curve γ . Then on a neighborhood of θ_0 the Riesz projection $P(\theta) = \frac{1}{2\pi i} \oint_\gamma (zI - A(\theta))^{-1} dz$ is real-analytic and rank-one, the eigenvalue $\lambda(\theta) = \operatorname{tr}(A(\theta)P(\theta))$ is real-analytic, and analytically normalized left/right eigenvectors exist. The reduced resolvent $S(\theta) = \frac{1}{2\pi i} \oint_\gamma \frac{(zI - A)^{-1}}{z - \lambda(\theta)} dz$ is real-analytic, and the first-order formulas $d\lambda = \operatorname{tr}(P dA)$ and $dP = -(S dA P + P dA S)$ hold.*

Applying Lemma B.1 to $A = Q$, $\lambda_0 = -\Lambda$, $S = R$ yields Proposition 2.6(i) and re-derives Theorem 4.1(a)–(b) intrinsically; the simplex-normalized eigenvector formula Theorem 4.1(c) is the additional content special to the probability normalization $\nu \mathbf{1} = 1$.

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